DIAL: Telling Invalid Instruments from Local Effects When IV and DML Disagree

Author names withheld

Editor: Under Review for CLeaR 2027

Abstract

When IV and DML estimates disagree, the cause is unclear: the instrument may be invalid, it may identify a local effect (LATE) that differs from the DML estimand, or DML may carry residual confounding. We develop DIAL (Diagnosing Instrument validity And Locality), a filter that checks relevance, validity, and then agreement between IV and DML. Its validity gate asks how often a Modified Conditional Union Bound confidence set for the instrument–error correlation contains zero across bootstrap resamples. We show that the gate checks whether the implied endogeneity r^*, the treatment–error correlation under which the data fit a valid instrument, lies in a domain that the researcher sets from the structural story. We derive finite-sample error bounds and a classifiability frontier for the gate. Given the domain, the test detects linear exclusion violations that the Hilbert-Schmidt Independence Criterion (HSIC) misses, and it accepts a valid heteroskedastic instrument that HSIC rejects. Across 13 instruments from eight published studies, 5 pass the validity gate and 8 fail it; IV and DML agree for one of the five.

Keywords: instrumental variables, double machine learning, partial identification, exclusion restriction, treatment-effect heterogeneity

1. Introduction

Causal inference in deployed machine-learning systems relies on two widely used approaches. Instrumental variables (IV) identify causal effects under unobserved confounding. They use an exogenous instrument that affects treatment but affects the outcome only through treatment (Imbens & Angrist, 1994). Double Machine Learning (DML) estimates effects under selection on observables, using flexible nuisance models with cross-fitting (Chernozhukov et al., 2018). The two approaches rest on non-nested identification assumptions: IV requires a valid instrument, while DML requires that observed covariates block all confounding. When both sets of assumptions hold and the effect is the same for everyone, the two estimates should agree. In practice, they often do not.

This diagnostic problem arises across machine-learning applications. Recommendation debiasing, off-policy evaluation, algorithmic fairness auditing, and treatment-effect estimation all rely on IV methods, often with a questionable exclusion restriction. Meanwhile, the growing literature on heterogeneous treatment effects (Wager & Athey, 2018; Künzel et al., 2019) makes the distinction between the Local Average Treatment Effect (LATE) and the Average Treatment Effect (ATE) more important, because the target estimand depends on the population that a decision serves. Existing tools study instrument validity or heterogeneity in isolation and do not tell the two explanations apart.

We propose DIAL (Diagnosing Instrument validity And Locality) to fill this gap. DIAL is a probabilistic diagnostic built on a random set: the Modified Conditional Union Bound (MCUB) confidence set $\hat{\mathcal{C}}_{\text{MCUB}}$ for the instrument–error correlation ρ_{ZU} (Bei, 2024). The validity verdict asks whether this set contains zero, and how reliably: DIAL recomputes the set on bootstrap resamples of the data and calls the instrument valid when at least a share $\tau_V = 0.95$ of the resampled sets contain

© 2027 .

zero. With one instrument, the model is just identified, so the data alone cannot test $\mathbb{E}[ZU] = 0$. The CR test supplies the missing information through a domain $\mathcal{D}$ for the treatment–error correlation, and we show that the verdict asks whether the implied endogeneity r^* lies in $\mathcal{D}$ (Proposition 4). We derive Hoeffding and exact-binomial bounds that give finite-sample classification guarantees, uniform over instruments, at every sample size n. Like a power curve, the classifiability frontier δ^* (Proposition 2) maps (n, B, π^*) to the minimum detectable violation, where B is the bootstrap size and π^* is the target error level.

Contributions. DIAL integrates the Correlation Restriction (CR) test (Dzhumashev & Tursunalieva, 2026) with Double Machine Learning (Chernozhukov et al., 2018). Using partial identification, the CR test answers *is the instrument invalid?* The IV/DML ratio answers *do IV and DML agree?* Section 3 formalises the full decision pipeline (Figure 1 and Algorithm 1). We make three contributions.

1. **Separable signatures and finite-sample guarantees:** an exclusion violation moves the validity verdict, while effect heterogeneity and residual DML confounding move only the IV/DML ratio (Theorem 1). Classification error bounds and a classifiability frontier follow (Proposition 2).

2. **What the validity gate checks:** the gate asks whether the implied endogeneity r^* lies in the maintained domain $\mathcal{D}$ for ρ_{DU}, and the sign of r^* fixes the side of OLS on which IV falls (Proposition 4). The domain is the prior information that makes one instrument testable; with it, the CR test detects invalid instruments that HSIC misses and accepts a valid heteroskedastic instrument that HSIC rejects (Corollary 3, Section 5.1).

3. **Empirical validation:** across 13 instruments in 8 applications, each under its sign-elicited canonical domain, 5 instruments pass the validity gate and 8 fail it, and no verdict changes under within-sign widening. IV and DML agree for one valid instrument and disagree for four, but bootstrap intervals for R show that these labels are less certain than the validity verdicts (Table 5). In Di Tella's data, two instruments for the same treatment receive verdicts A and B.

2. Background, Related Work, and Positioning

IV estimation and the LATE-ATE distinction. In the linear triangular model $Y = \beta D + U$, $D = \pi Z + \varepsilon$, IV identifies β_{LATE} for compliers (Imbens & Angrist, 1994), while DML identifies θ_{DML} under selection on observables (Chernozhukov et al., 2018). With homogeneous effects, $\theta_{\text{DML}} = \beta_{\text{ATE}}$. With heterogeneous effects, the partially linear DML estimand is a variance-weighted average of the individual effects, and $\beta_{\text{LATE}} \neq \theta_{\text{DML}}$ even with a valid instrument (Heckman & Vytlacil, 2005).

The Correlation Restriction test. The CR test (Dzhumashev & Tursunalieva, 2026) uses partial identification (Manski, 2003; Tamer, 2010) to assess whether the exclusion restriction holds. It takes the observable correlations $\boldsymbol{\rho} = (\rho_{DY}, \rho_{DZ}, \rho_{YZ})'$ and a maintained domain $\mathcal{D} \subset (-1, 1)$ for the unobserved endogeneity ρ_{DU}. Substituting $\rho_{ZU} = \text{Cov}(Z, U)/(\text{sd}(Z)\,\text{sd}(U))$ into the structural relation $Y = \beta D + U$ yields the CR mapping (DiTraglia & García-Jimeno, 2021)

$$g(\boldsymbol{\rho}, r) = \rho_{DZ}\, r - (\rho_{DY}\rho_{DZ} - \rho_{YZ})\sqrt{\frac{1 - r^2}{1 - \rho_{DY}^2}}, \tag{1}$$

whose image $\mathcal{C} = \{g(\boldsymbol{\rho}, r) : r \in \mathcal{D}\}$ is an identified set for ρ_{ZU} (derivation in Dzhumashev & Tursunalieva (2026), Appendix A). Under heterogeneous effects, we define $U := Y - \beta_{\text{LATE}} D$, net of covariates. A valid instrument then keeps $\rho_{ZU} = 0$, and ρ_{DU} absorbs both the confounding and the effect variation $(\beta_i - \beta_{\text{LATE}})D$ (Proposition 4). In practice, the researcher elicits the sign of ρ_{DU} from the structural story, which restricts $\mathcal{D}$ to one half of $(-1, 1)$, and then narrows $\mathcal{D}$ to a canonical subinterval. Corollary 3(c) formalises the failure mode of a misspecified sign. If the instrument is valid, then $0 \in \mathcal{C}$. We base inference on the MCUB (Bei, 2024), which combines the Imbens–Manski projection (Imbens & Manski, 2004) with a Bonferroni adjustment over the domain grid (Stoye, 2009). By construction, $\Pr\big(\mathcal{C} \subseteq \hat{\mathcal{C}}_{\text{MCUB}}\big) \geq 1 - \alpha_0$ at nominal level α_0. The first coordinate of DIAL, the validity verdict V, checks whether $\hat{\mathcal{C}}_{\text{MCUB}}$ contains zero across bootstrap resamples (Section 4.3).

IV validity testing and independence-based diagnostics. Pearl (1995) conjectured that instrument validity cannot be tested with continuous treatments; Gunsilius (2021) showed that mild assumptions restore partial testability. Classical overidentification tests (Sargan, 1958; Hansen, 1982) require a second instrument, while sensitivity analyses (Conley et al., 2012; Cinelli & Hazlett, 2020) bound the bias under violations that the researcher specifies. Independence-based diagnostics test the strictly stronger condition $Z \perp\!\!\!\perp U$, which also rules out higher-order dependence beyond $\rho_{ZU} = 0$. They include IV-GIN (Instrumental Variable Generalized Independent Noise; Xie et al. 2020, 2022), KCI (Kernel Conditional Independence test; Zhang et al. 2011), and HSIC (Hilbert-Schmidt Independence Criterion; Gretton et al. 2008). In non-Gaussian settings, a valid instrument can satisfy $\rho_{ZU} = 0$ yet fail these tests. The CR test instead gives a binary verdict on $\rho_{ZU} = 0$ from a single just-identified instrument, with the domain $\mathcal{D}$ as prior information. Section 5.1 shows that, given $\mathcal{D}$, it detects exclusion violations that HSIC misses, and it admits a valid heteroskedastic instrument that HSIC rejects.

Heterogeneous treatment effects. Causal forests (Wager & Athey, 2018), metalearners (Künzel et al., 2019), and doubly robust methods (Kennedy, 2022; Foster & Syrgkanis, 2023) estimate heterogeneous effects, and sensitivity analysis (Yadlowsky et al., 2022) and causal representation learning (Schölkopf et al., 2021) address related problems. Oprescu & Kallus (2024) combine a weak IV with observational data through a compliance-weighted correction to estimate conditional average treatment effects (CATEs) when compliance is low. DIAL has a different goal. Rather than fuse estimators, it diagnoses *why* IV and DML disagree and gives finite-sample classification guarantees. To our knowledge, no existing method distinguishes the three cases jointly and probabilistically with finite-sample guarantees. Table 2 (Appendix A) collects the notation.

3. The DIAL Framework

3.1. Diagnostic Algorithm

DIAL computes two quantities: the validity verdict $V = \mathbf{1}\{\widehat{p}_B \geq \tau_V\}$, where $\widehat{p}_B$ is the share of B bootstrap resamples whose MCUB set contains zero, and the IV/DML ratio $R = |\hat{\beta}_{\text{IV}}|/|\hat{\beta}_{\text{DML}}|$. Classification follows Algorithm 1, a sequential filter with three stages: *relevance* (Stock–Yogo), then *validity* (CR via MCUB), then *agreement* (ratio and signs). A stage can be interpreted only after the stage before it has passed, so the algorithm stops at the first failed stage and returns A, B, or C.

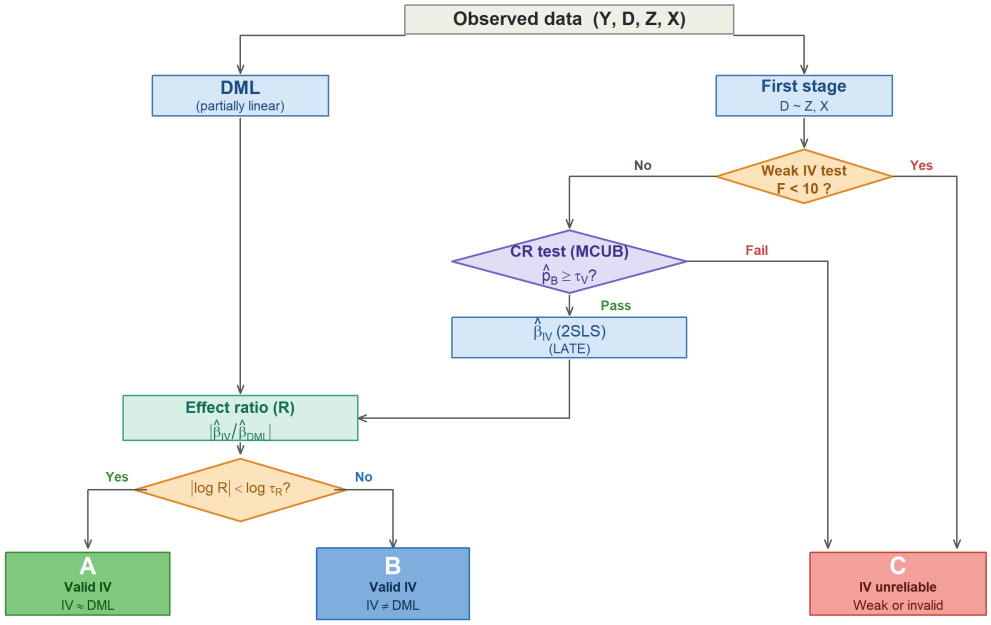


Figure 1: The DIAL diagnostic framework. The right leg runs the first stage ($D \sim Z, X$); an instrument with $F < 10$ ends in Scenario C. An instrument that passes this relevance gate goes to the CR test (MCUB), whose bootstrap rule asks *is the instrument invalid?* An invalid instrument ends in Scenario C, and a valid one goes on to 2SLS. The left leg computes $\hat{\beta}_{\text{DML}}$, the selection-on-observables benchmark and fallback. The ratio $R = |\hat{\beta}_{\text{IV}}/\hat{\beta}_{\text{DML}}|$ then asks *do IV and DML agree?* Scenario A requires $|\log R| < \log \tau_R$ and estimates of the same sign; otherwise, DIAL returns Scenario B.

3.2. Diagnostic Space and Core Insight

In the model $Y = \beta D + \gamma Z + U$, three mechanisms can drive $\hat{\beta}_{\text{IV}} \neq \hat{\beta}_{\text{DML}}$. An exclusion violation ($\gamma \neq 0$) induces $\rho_{ZU} \neq 0$, shifts the identified set $\mathcal{C}$ *away from zero*, flips the validity verdict V to 0, and biases the IV estimand. Treatment-effect heterogeneity (individual effects β_i) re-weights compliers but creates no $Z \to Y$ path: with $U := Y - \beta_{\text{LATE}} D$, a valid instrument keeps $\rho_{ZU} = 0$ and $V = 1$, while IV targets β_{LATE} and DML the variance-weighted θ_{DML}. Confounding that the covariates miss biases DML but leaves V unchanged. All three mechanisms can move R away from 1, so R alone cannot tell them apart, and V alone cannot tell A from B. Together, (V, R) partition the diagnostic space into three exclusive scenarios, with $\tau_R := 1.5$ and $\tau_V := 0.95$ (Section 4). In Scenario A (valid; IV and DML agree), $V = 1$, the estimates share a sign, and $1/\tau_R < R < \tau_R$. In Scenario B (valid; IV and DML disagree), $V = 1$, and R falls outside this band or the signs differ: IV identifies a LATE that differs from θ_{DML}, or DML carries residual confounding, and one instrument cannot tell which (Proposition 4). In Scenario C (unreliable), $V = 0$ or the first stage is weak, and the analyst falls back to DML; DIAL records the subtype, *weak* or *invalid*.

Algorithm 1 DIAL: diagnosing instrument validity and locality. Three-stage filter: relevance → validity → agreement.

Require: *Data:* instrument Z, treatment D, outcome Y, covariates X
Require: *Researcher inputs:* maintained domain $\mathcal{D} \subset (-1, 1)$ for ρ_{DU} (sign-elicited per Section 4.4, step **(1)**), ratio threshold τ_R (default 1.5; formalised in Assumption 4, Section 4.1)
Require: *Thresholds:* $B = 500$ resamples, coverage $\tau_V = 0.95$ (Section 4.3)

{Stage 1: Relevance}
Estimate $\hat{\beta}_{\text{DML}}$ via cross-fitted nuisance regression on (Y, D, X)
Compute first-stage F-statistic from regressing D on (Z, X)
if $F < 10$ **then**
  return C (weak): IV unreliable; use $\hat{\beta}_{\text{DML}}$
end if
{Stage 2: Validity}
Compute the MCUB set $\hat{\mathcal{C}}_{\text{MCUB}}$ for ρ_{ZU} over $\mathcal{D}$ (Section 2), and recompute it on each of B bootstrap resamples; let $\widehat{p}_B$ be the share of resampled sets that contain zero (within tolerance; footnote 2)
if $\widehat{p}_B < \tau_V$ **then**
  return C (invalid): exclusion violation detected; use $\hat{\beta}_{\text{DML}}$
end if
{Stage 3: Agreement of IV and DML}
Estimate $\hat{\beta}_{\text{IV}}$ via 2SLS using (Z, D, Y, X); set $R \leftarrow |\hat{\beta}_{\text{IV}}| \,/\, |\hat{\beta}_{\text{DML}}|$ and compute r^* (Proposition 4)
if $\hat{\beta}_{\text{IV}}\, \hat{\beta}_{\text{DML}} > 0$ **and** $|\log R| < \log \tau_R$ **then**
  return A: valid IV; IV and DML agree; either estimate usable
else
  return B: valid IV; IV and DML disagree (local effect or residual DML confounding); report r^* and R with bootstrap intervals; label B (sign) if the signs differ
end if

4. Theoretical Foundations: Separation, Frontier, Domain

Four results formalise the partition of Section 3.2: a separation theorem (Theorem 1), finite-sample error bounds (Proposition 2), a domain-sensitivity result (Corollary 3), and an identity for what the validity gate checks (Proposition 4).

Notation. Let $\mathcal{D} \subset (-1, 1)$ denote the maintained domain for ρ_{DU}, which we assume contains the true ρ_{DU}. Define the exclusion-violation magnitude $\delta := |\rho_{ZU}|$ and the MCUB coverage probability

$$\text{CP}_n(\delta) := \Pr_{\delta}\big(0 \in \hat{\mathcal{C}}_{\text{MCUB}}\big).$$

Let $\widehat{p}_B$ be the share of B bootstrap resamples whose recomputed MCUB set contains zero, and $V_n := \mathbf{1}\{\widehat{p}_B \geq \tau_V\}$ the validity verdict of Algorithm 1.

4.1. Assumptions

Assumption 1 (Coverage monotonicity) *The map $\delta \mapsto \text{CP}_n(\delta)$ is continuous and strictly decreasing on its domain.*

Assumption 2 (Estimator consistency on the valid region) *Conditional on* $\{F \geq 10\}$ *and* $\delta = 0$, $\sqrt{n}(\hat{\beta}_{IV} - \beta_{LATE}) = O_p(1)$ *and* $\sqrt{n}(\hat{\beta}_{DML} - \theta_{DML}) = O_p(1)$, *where* θ_{DML} *is the DML estimand, and there exists* $b_0 > 0$ *such that* $|\beta_{LATE}| \geq b_0$ *and* $|\theta_{DML}| \geq b_0$. *Hence* $R_n := |\hat{\beta}_{IV}|/|\hat{\beta}_{DML}| \to_p R_0 := |\beta_{LATE}|/|\theta_{DML}|$ *and* $S_n := \operatorname{sign}(\hat{\beta}_{IV}\hat{\beta}_{DML}) \to_p s_0 := \operatorname{sign}(\beta_{LATE}\,\theta_{DML})$.

Here $\theta_{\text{DML}} = \beta_{\text{ATE}}$ when effects are homogeneous and the covariates block all confounding; b_0 keeps both estimands away from zero.

Assumption 3 (MCUB classifiability) *For each target error level* $\pi^* \in (0, 1)$ *and bootstrap size* $B \in \mathbb{N}$, *there exists a classifiability frontier* $\delta^* = \delta^*(n, B, \sigma, \pi^*) > 0$ *such that, for every finite* n,

$$\Pr_{\delta=0}\big(V_n = 1\big) \geq 1 - \pi^*, \qquad \Pr_{\delta \geq \delta^*}\big(V_n = 0\big) \geq 1 - \pi^*.$$

Assumption 3 is the finite-sample detectability condition. Instruments with $0 < \delta < \delta^*$ lie in an unavoidable grey region near the detection boundary.

Assumption 4 (Ratio margin on the valid region) *Let* $\tau_R := 1.5$. *On* $\{\delta = 0\}$, *call the population* in agreement *(*A^**) if* $s_0 = 1$ *and* $|\log R_0| < \log \tau_R$, *and* in disagreement *(*B^**) otherwise. If* $s_0 = 1$, *there exists* $\eta > 0$ *such that* $\big|\,|\log R_0| - \log \tau_R\big| \geq \eta$.

The margin η keeps R_0 away from τ_R and $1/\tau_R$. Homogeneous effects with an unconfounded DML give $R_0 = 1$, hence A^*; heterogeneity or residual confounding gives B^* once it moves R_0 outside $(1/\tau_R, \tau_R)$.

4.2. Separation Theorem

Theorem 1 (Separation Theorem) *Under Assumptions 1–4 and conditional on* $\{F \geq 10\}$ *and on* $\mathcal{D}$ *containing the true* ρ_{DU}, *take the validity verdict* V_n *defined above, the ratio* R_n *and sign* S_n *as in Assumption 2,* τ_R *as in Assumption 4, and the* DIAL *decision regions*

$$A_n := \{V_n = 1,\, S_n = 1,\, |\log R_n| < \log \tau_R\}, \quad B_n := \{V_n = 1\} \setminus A_n, \quad C_n := \{V_n = 0\}.$$

Then:

1. ***(Partition, finite-sample.)*** A_n, B_n, C_n *are mutually exclusive and exhaustive on* $\{F \geq 10\}$.
2. ***(Validity separation, finite-sample.)*** *For every finite* n, $\Pr_{\delta=0}(A_n \cup B_n) \geq 1 - \pi^*$ *and* $\Pr_{\delta \geq \delta^*}(C_n) \geq 1 - \pi^*$.
3. ***(Agreement separation, asymptotic.)*** *As* $n \to \infty$, $\Pr(A_n \mid \delta = 0,\, \mathrm{A}^*) \geq 1 - \pi^* - o(1)$ *and* $\Pr(B_n \mid \delta = 0,\, \mathrm{B}^*) \geq 1 - \pi^* - o(1)$.

If $\mathcal{D}$ *does not contain the true* ρ_{DU}, *the guarantees in Parts 2 and 3 can break down.*

Appendix C.1 gives a proof sketch and the full proof.

4.3. Finite-Sample Theory

Assumption 3 is a high-level abstraction; the next proposition makes it concrete through error bounds for the bootstrap validity rule.

Bootstrap coverage threshold. Let $\tau_V \in (0,1)$ denote the bootstrap coverage threshold. DIAL declares an instrument *Valid* when the fraction of bootstrap resamples with $0 \in \hat{\mathcal{C}}^b_{\text{MCUB}}$ is at least τ_V, and *Invalid* otherwise.[1] Each resample redraws the rows of the data and recomputes the whole MCUB set, data preparation and covariance bootstrap included (Appendix G). We fix $\tau_V = 0.95$ and $B = 500$ in all experiments. The Monte Carlo results (Figure 2, Table 7) report the single-sample coverage $\text{CP}_n(\delta)$ instead, because the rule would multiply their cost by B; Table 8 runs the rule itself on three cells. The threshold τ_V separates valid from invalid instruments (Scenarios A∪B vs. C), while τ_R separates agreement from disagreement of IV and DML (Scenarios A vs. B).

Proposition 2 (Finite-sample bootstrap classification error) *Let $I_b := \mathbf{1}\{0 \in \hat{\mathcal{C}}^b_{MCUB}\}$ for $b = 1, \ldots, B$ be bootstrap validity indicators with conditional success probability $p^\star_n := \Pr^\star(I_1 = 1 \mid \mathcal{X}_n)$, and let $\widehat{p}_B := B^{-1}\sum_b I_b$. Classify an instrument* Valid *when $\widehat{p}_B \geq \tau_V$,* Invalid *otherwise.*

*(a) (*Conditional Hoeffding bound.*) For $p^\star_n \neq \tau_V$, $\Pr^\star(E_{MC} \mid \mathcal{X}_n) \leq \exp(-2B|p^\star_n - \tau_V|^2)$, where E_{MC} is the relevant misclassification event.*

*(b) (*Frontier sufficient condition.*) Set $\epsilon_H(\pi^*) := \sqrt{\log(1/\pi^*)/(2B)}$. If there exists δ^*_H such that $\Pr_\delta(p^\star_n \leq \tau_V - \epsilon_H(\pi^*)) \geq 1 - \alpha_n(\delta)$ for all $\delta \geq \delta^*_H$, then $\Pr_\delta($classify as Invalid$) \geq 1 - \pi^* - \alpha_n(\delta)$. The valid side ($\delta = 0$) is symmetric, with the exact binomial threshold $p_{\min}$ in place of $\tau_V + \epsilon_H(\pi^*)$, which exceeds one at our B and τ_V ($p_{\min} = 0.963$ at $\pi^* = 0.05$).*

*(c) (*Closed form.*) If $p^\star_n = m_n(\delta)$ almost surely for a continuous strictly decreasing m_n, then $\delta^*_H(n, B, \pi^*) = m_n^{-1}(\tau_V - \epsilon_H(\pi^*))$.*

Appendix C.2 gives exact binomial probabilities for Part (a), a proof sketch, and the full proof.

Interpretation and detection thresholds. Part (a) is conditional on the sample, Part (b) gives the repeated-sample guarantee, and Part (c) gives the operational formula. For order-of-magnitude calculations, Remark 5 (Appendix C.2) adds a Gaussian closed form. Under partial identification, this closed form governs the distance $\kappa = d\big(0, C(\rho, \overline{\mathcal{D}})\big)$ between zero and the identified set (Corollary 3(b)), not $|\rho_{ZU}|$ itself. With $\sigma = 3.25$, the median value calibrated on the eight invalid empirical instruments (Appendix D.3), DIAL resolves a set whose nearest endpoint lies 0.015 from zero at $n = 10{,}000$ and 0.044 at $n = 1{,}107$ (Table 6). A violation therefore escapes detection at every n as long as some $r \in \mathcal{D}$ still rationalises the data with a valid instrument, because then $\kappa = 0$.

Figure 2 plots the power curves $\text{CP}_n(\delta)$ for five sample sizes. In its design, the identified set contains zero whenever $\delta \leq \rho_{DU}\,\text{cor}(Z, D) = 0.125$. This value, δ^*_∞, is the detection threshold as $n \to \infty$. The crossing δ^*_n, read from the dashed verticals in panel (b), falls from 0.170 at $n = 1{,}107$ to 0.131 at $n = 50{,}000$. Its excess over δ^*_∞ falls from 0.045 to 0.006, roughly at the $1/\sqrt{n}$ rate that the MCUB widths in panel (c) also show. The grey region of Assumption 3 lies between the plateau, where $\text{CP}_n(\delta) \geq 1 - \pi^*$, and the region where $\text{CP}_n(\delta) \leq \pi^*$.

4.4. Sensitivity to the Maintained Domain

The results so far assume that $\mathcal{D}$ contains the true ρ_{DU}. The MCUB set itself depends on the domain, so widening or misspecifying $\mathcal{D}$ can substantially change coverage and, with it, the DIAL classification.

1. The naive percentile bootstrap can be inconsistent for partially identified parameters (Bugni, 2010; Romano et al., 2014). MCUB avoids this problem: it bootstraps the correlations through a continuous map and recovers the set-valued parameter with delta-method arguments. Lemma 6 (Appendix C.3) establishes Hausdorff consistency.

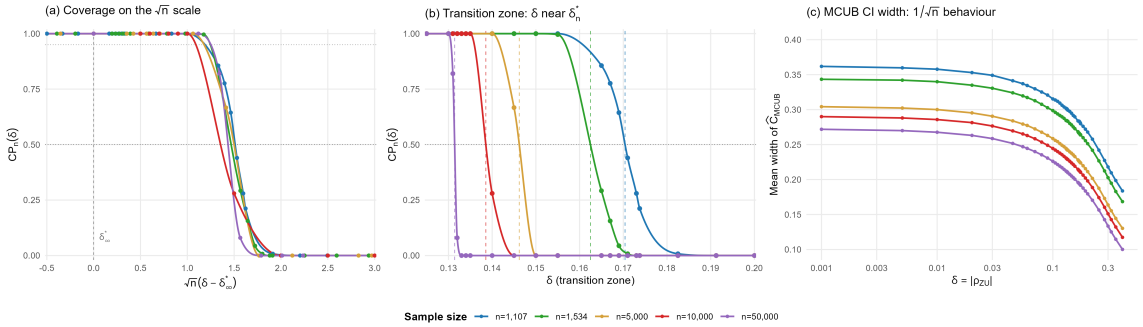


Figure 2: Power curves $\mathrm{CP}_n(\delta) = \Pr\big(0 \in \hat{\mathcal{C}}_{\mathrm{MCUB}}\big)$ and the classifiability frontier for $n \in \{1{,}107,\ 1{,}534,\ 5{,}000,\ 10{,}000,\ 50{,}000\}$. The Gaussian design sets $\rho_{DU} = 0.5$, $\mathrm{cor}(Z, D) = 0.25$, and $\mathcal{D} = (0, 0.8)$, and uses the strict single-sample verdict at $\alpha_0 = 0.05$ (settings in Appendix G). **(a)** Against $\sqrt{n}\,(\delta - \delta^*_\infty)$, the curves nearly coincide, as Remark 5 predicts. Dotted lines mark $\mathrm{CP} = 0.95$ and 0.5. **(b)** Monotone cubic curves through the Monte Carlo estimates (points) for $\delta \in [0.125, 0.200]$; dashed lines mark δ^*_n, where a curve crosses 0.5. **(c)** Mean width of $\hat{\mathcal{C}}_{\mathrm{MCUB}}$; at fixed δ, the gap between sample sizes is the sampling part of the width.

Corollary 3 (Sign and domain sensitivity) *Let $\mathcal{D} \subset (-1, 1)$ be the working domain, $C(\rho, \overline{\mathcal{D}}) := \{g(\rho, r) : r \in \overline{\mathcal{D}}\}$ the identified set $\mathcal{C}$ of Section 2, taken over $\overline{\mathcal{D}}$, and $CP_n(\delta, \mathcal{D}) := \Pr_\delta(0 \in \hat{\mathcal{C}}_{MCUB,n}(\mathcal{D}))$ the coverage probability. Assume the MCUB set is almost surely nonempty and compact, that it is monotone in $\mathcal{D}$, and that it is Hausdorff-consistent for $C(\rho, \overline{\mathcal{D}})$. Then:*

(a) $\mathcal{D}_1 \subset \mathcal{D}_2 \implies CP_n(\delta, \mathcal{D}_2) \geq CP_n(\delta, \mathcal{D}_1)$.

(b) If $0 \notin C(\rho, \overline{\mathcal{D}})$ *(equivalently* $\kappa(\mathcal{D}) := d(0, C(\rho, \overline{\mathcal{D}})) > 0$*), then* $CP_n(\delta, \mathcal{D}) \to 0$.

(c) In particular, if the instrument is valid ($\delta = 0$) and $r \mapsto g(\rho, r)$ has a unique zero at $r = \rho_{DU}$, then $\rho_{DU} \notin \overline{\mathcal{D}} \implies CP_n(0, \mathcal{D}) \to 0$: misspecifying the sign or excluding the true ρ_{DU} collapses coverage to zero even for valid instruments.

Appendix C.3 gives the full proof and links $\mathcal{D}$ to credal priors (Remark 8).

The next result says what the validity gate checks.

Proposition 4 (Implied endogeneity) *Let $\rho_{DZ} \neq 0$, $|\rho_{DY}| < 1$, and $k := (\rho_{DY}\rho_{DZ} - \rho_{YZ})/(\rho_{DZ}\sqrt{1 - \rho^2_{DY}})$.*

(a) The map $r \mapsto g(\boldsymbol{\rho}, r)$ has exactly one zero in $(-1, 1)$, at $r^* := k/\sqrt{1 + k^2}$. Hence $0 \in C(\rho, \overline{\mathcal{D}})$ if and only if $r^* \in \overline{\mathcal{D}}$.

(b) $r^* = \mathrm{cor}(D, Y - \beta_{IV} D)$ and $\beta_{OLS} - \beta_{IV} = r^* \,\mathrm{sd}(Y - \beta_{IV} D)/\mathrm{sd}(D)$, where β_{IV} and β_{OLS} are the IV and OLS slopes. So $\mathrm{sign}(r^*) = \mathrm{sign}(\beta_{OLS} - \beta_{IV})$.

(c) If Z is valid and $U := Y - \beta_{LATE} D$, then $\rho_{ZU} = 0$ and $r^* = \rho_{DU}$, which mixes the confounding in U with the effect variation $(\beta_i - \beta_{LATE})D$.

All variables are net of the covariates X, and (a) and (b) also hold for sample moments. Appendix C.4 gives the proof.

By (a), the validity gate asks whether the implied endogeneity r^*, the value of ρ_{DU} under which the data fit a valid instrument, lies in $\mathcal{D}$. The domain thus carries the prior information that a just-identified model lacks, and the CR test's advantage over HSIC (Section 5.1) comes from it. By (b), the sign of r^* fixes the side of OLS on which IV falls: under a positive domain and a positive effect, a valid instrument has $\beta_{\text{IV}} < \beta_{\text{OLS}}$, so $R < 1$ when DML is close to OLS, and the ratio rule must be two-sided. By (c), a local effect and residual confounding enter r^* in the same way, so Scenario B cannot attribute the IV–DML gap to either one; r^* measures its size in correlation units.

Practical protocol. We recommend a four-step protocol for choosing and validating $\mathcal{D}$; Algorithm 1, Table 4, and Figure 3 refer to it.

(1) *Sign elicitation.* Elicit the sign of ρ_{DU} from the structural story, and commit to projecting over either $(0, 1)$ or $(-1, 0)$. Corollary 3(c) formalises the failure mode of sign misspecification, that is, projecting over the wrong half.

(2) *Canonical subinterval.* Choose a canonical subinterval of the elicited half-line as the primary domain. For example, use $(0, 0.8)$ for positive endogeneity when the structural story pins ρ_{DU} down only loosely, and a narrower interval when the story pins it down tightly.

(3) *Within-sign robustness check.* Widen $\mathcal{D}$, within the same half-line, to the widest structurally admissible subinterval; we call this the robustness domain. An instrument is *robustly valid* or *robustly invalid* if its validity verdict V_n is the same under the canonical and the robustness domains. Extending the check across the sign boundary (e.g. to $(-0.5, 0.5)$ after a positive commitment) is *not* a robustness check. It re-admits $\rho_{DU} = 0$, which, as Corollary 3(c) shows, can collapse coverage even for a valid instrument.

(4) *Implied endogeneity.* Report r^* with a bootstrap interval. An interval that reaches the edge of $\mathcal{D}$ flags a verdict that rests on where that edge sits.

Section 5 tests the empirical content of these results: against HSIC (§5.1), across all three scenarios (§5.2), and on 13 instruments (§5.3).

5. Experiments: Synthetic, Structural, and Empirical

Replication code for every table and figure is at https://anonymous.4open.science/r/dial-cr-replication-0B38.

5.1. Synthetic Benchmark: CR Test vs. HSIC under the Jones-Pewsey DGP

We use the non-Gaussian Jones-Pewsey DGP of Appendix D.1. The regressor X is the sum of two independent sinh-arcsinh-transformed normals (Jones & Pewsey, 2026). The structural error $U = \lambda(X - \bar{X}) + \varepsilon + v$ has an endogenous component controlled by λ and an exogenous component v orthogonal to X, and the outcome is $Y = 2X + U$. Here X plays the role of the treatment D, and there are no covariates. We calibrate $|\lambda|$ so that $|\rho_{XU}| \approx 0.30$, and we construct eleven instruments at known $\rho_{ZU} \in \{-0.20, -0.16, \ldots, +0.20\}$ with matched first-stage strength $\text{cor}(Z, X) = 0.25$. A twelfth instrument, Z_{het}, is valid but heteroskedastic: its error satisfies $\mathbb{E}[U \mid Z] = 0$, while

$\mathrm{Var}(U \mid Z)$ changes with Z (Appendix D.1). Following the protocol of Section 4.4, we draw the instruments with $\rho_{ZU} < 0$ from the $\lambda < 0$ DGP and test them under $\mathcal{D}^- = (-0.8, 0)$; the others use $\lambda > 0$ and $\mathcal{D}^+ = (0, 0.8)$. We use $n = 800$, $\alpha_0 = 0.10$, and a finite-sample equivalence margin $\mathrm{ZEROTOL} = 0.05$ on the MCUB endpoints.[2] We test HSIC with its gamma approximation at the 5% level.

Table 1: Synthetic benchmark under the Jones-Pewsey DGP. Each row uses its correctly elicited domain $\mathcal{D}$ (Section 4.4). $\widehat{p}_B$ is the share of bootstrap resamples whose MCUB set contains zero within ZEROTOL. Settings: $n = 800$, $\alpha_0 = 0.10$, $\mathrm{ZEROTOL} = 0.05$; HSIC at the 5% level.

Instrument	ρ_{ZU}	$\mathcal{D}$	$\hat{\beta}_{\mathrm{IV}}$	$\hat{\mathcal{C}}_{\mathrm{MCUB}}$	$\widehat{p}_B$	HSIC p	DIAL	HSIC agrees?
$Z_{-0.20}$	-0.20	$(-0.8, 0)$	1.027	$[-1.000, -0.265]^{\dagger}$	0.032	0.280	C	No
$Z_{-0.16}$	-0.16	$(-0.8, 0)$	1.222	$[-1.000, -0.116]^{\dagger}$	0.282	0.477	C	No
$Z_{-0.12}$	-0.12	$(-0.8, 0)$	1.416	$[-0.998, +0.060]$	0.728	0.131	C	No
$Z_{-0.08}$	-0.08	$(-0.8, 0)$	1.611	$[-0.948, +0.235]$	0.974	0.098	A	Yes
$Z_{-0.04}$	-0.04	$(-0.8, 0)$	1.805	$[-0.859, +0.393]$	1.000	0.348	A	Yes
$Z_{+0.00}$	0.00	$(0, 0.8)$	2.000	$[-0.499, +0.771]$	1.000	0.883	A	Yes
$Z_{+0.04}$	$+0.04$	$(0, 0.8)$	2.184	$[-0.340, +0.861]$	0.998	0.170	A	Yes
$Z_{+0.08}$	$+0.08$	$(0, 0.8)$	2.367	$[-0.200, +0.955]$	0.960	0.895	A	Yes
$Z_{+0.12}$	$+0.12$	$(0, 0.8)$	2.551	$[-0.017, +0.999]$	0.700	0.007	C	Yes
$Z_{+0.16}$	$+0.16$	$(0, 0.8)$	2.735	$[+0.138, +1.000]^{\dagger}$	0.242	0.165	C	No
$Z_{+0.20}$	$+0.20$	$(0, 0.8)$	2.918	$[+0.295, +1.000]^{\dagger}$	0.022	0.690	C	No
Z_{het}	0.00	$(0, 0.8)$	2.325	$[-0.271, +1.000]^{\dagger}$	0.950	< 0.001	A	No

† MCUB endpoints of -1.034 ($Z_{-0.20}$), -1.023 ($Z_{-0.16}$), $+1.025$ ($Z_{+0.16}$), $+1.033$ ($Z_{+0.20}$), and $+1.004$ (Z_{het}) are clipped to $[-1, 1]$; the verdicts are unaffected.

Table 1 shows how the CR test and HSIC differ. HSIC tests $Z \perp\!\!\!\perp \hat{e}$, where $\hat{e} = Y - \hat{\beta}_{\mathrm{IV}} X$. This condition is stronger than $\mathbb{E}[ZU] = 0$, yet $\hat{e}$ is uncorrelated with Z by construction, so HSIC cannot see the linear leakage that the IV estimator absorbs into $\hat{\beta}_{\mathrm{IV}}$. As a result, HSIC fails to reject five of the six invalid instruments, including all four at $|\rho_{ZU}| \in \{0.16, 0.20\}$, while DIAL rejects all six and accepts all five instruments with $|\rho_{ZU}| \leq 0.08$. This advantage comes from the prior information in $\mathcal{D}$, not from extra power in the data: by Proposition 4(a), the CR test rejects when r^* falls outside $\mathcal{D}$. With $\mathcal{D} = (-1, 1)$, its identified set contains zero for any relevant instrument, since $g(\boldsymbol{\rho}, \pm 1) = \pm \rho_{DZ}$. At $|\rho_{ZU}| = 0.12$, the MCUB sets still contain zero, but only 73% and 70% of the resampled sets do, so the bootstrap rule rejects both instruments. The last row shows the reverse error: HSIC rejects the valid instrument Z_{het} ($p < 0.001$) because $\mathrm{Var}(U \mid Z)$ depends on Z, while DIAL accepts it, narrowly ($\widehat{p}_B = 0.950 = \tau_V$).

2. Strict containment rejects valid instruments whose MCUB endpoints sit a sampling-noise distance, $O(n^{-1/2})$, from zero. We therefore check $\mathbf{1}\{\hat{\ell} \leq \mathrm{ZEROTOL}\} \cdot \mathbf{1}\{\hat{u} \geq -\mathrm{ZEROTOL}\}$ on $[\hat{\ell}, \hat{u}] = \hat{\mathcal{C}}_{\mathrm{MCUB}}$, for the full-sample set and for every resampled set. Tables 1, 7, and 9 use this margin; the empirical applications use strict containment at $\alpha_0 = 0.05$ (Tables 3 and 4). In Table 1, the margin changes no full-sample verdict: each set either contains zero or lies at least 0.116 from it.

5.2. Structural Simulation: All Three Scenarios

One structural model reaches every scenario. It has two latent types with equal probability, $Y = \beta_i D + \gamma Z + \boldsymbol{\phi}^\top H + U$ and $D = \pi_i Z + \boldsymbol{\lambda}^\top H + V$, where $H \sim N(0, I_5)$ and the errors U and V share a confounder C (Appendix D). Each row draws its own sample of $n = 50{,}000$ and fits its own DML, with $\alpha_0 = 0.10$, $\mathcal{D} = (0, 0.8)$, and ZEROTOL $= 0.05$. Row A has a homogeneous effect $\beta = 0.425$. In row B (het.), the types have effects 0.80 and 0.05 and first stages 0.05 and 0.60. The low-effect type complies more, so IV targets $\beta_{\text{LATE}} = 0.108$, while DML targets the variance-weighted $\theta_{\text{DML}} = 0.301$. Row B (conf.) reuses the sample of row A but hides C, so DML targets 0.700 while IV still targets 0.425. Row C sets $\gamma = 0.30$, and row C (weak) uses a first stage that keeps F below 10.

Table 9 (Appendix F) confirms that each row lands in its predicted cell. The two B rows reach the same verdict by different routes, yet both have $r^* > 0$, inside $\mathcal{D}$: $U = Y - \beta_{\text{LATE}} D$ absorbs the effect variation in row B (het.) and the confounding through C in row B (conf.). As Proposition 4(c) predicts, the data cannot tell the two routes apart. The exclusion violation shifts $\hat{\mathcal{C}}_{\text{MCUB}}$ away from zero, and the weak instrument stops at the relevance gate.

5.3. Empirical Applications

We apply DIAL to 13 instruments across 8 published applications: criminal recidivism (Di Tella & Schargrodsky, 2013); recommender systems (MovieLens; Harper & Konstan, 2015); military service and crime (Galiani et al., 2011); retirement savings (Papke, 1995); nutritional epidemiology (NHANES[3]); returns to education (Card, 1995); school construction (Burde & Linden, 2013); and colonial land tenure (Banerjee & Iyer, 2005). We analyse each instrument under its sign-elicited canonical domain (Section 4.4) and use within-sign widening as the robustness check. Of the 13 instruments, 5 pass the validity gate (Scenarios A or B) and 8 fail it (Scenario C), and no verdict changes under within-sign widening (Table 4). In both domains, $\widehat{p}_B \geq 0.958$ for the five valid instruments and $\widehat{p}_B \leq 0.814$ for the eight invalid ones. Only for Card does the rule overturn the single-set verdict: its MCUB set contains zero, but only 81% of its resampled sets do. Two of the eight, MovieLens `release_year` and `is_sequel`, also fail the relevance gate at the item level ($F < 2$); every other instrument has a first-stage $F \geq 19$. Figure 4 (Appendix F) maps all 13 instruments, and Appendix B gives the numbers and discusses each one.

Among the five valid instruments, only Di Tella's `percJudgeSentToEM` has IV and DML in agreement (Scenario A). Three others land in Scenario B because R lies outside $(1/\tau_R, \tau_R)$, and MovieLens `num_genres` because its estimates have opposite signs. The ratio is much less precise than the validity verdict: the share of resamples that land in Scenario B ranges from 0.675 for `percJudgeSentToEM` to 0.995 for `num_genres` (Table 5). Every invalid instrument has r^* outside its domain; for Card, $r^* = -0.067$ lies just below it. Burde and Linden's instrument fails although school placement was randomised: randomisation rules out confounded placement, but a village school can raise test scores through channels other than enrolment (Appendix B.3).

The Di Tella showcase: same study, two scenarios. In the study of Di Tella & Schargrodsky (2013), two instruments for electronic-monitoring (EM) assignment land in different scenarios, although the outcome, treatment, and covariates are identical. `judgeAlreadyUsedEM` passes the validity gate, but IV and DML disagree ($\hat{\beta}_{\text{IV}} = -0.337$, $\hat{\beta}_{\text{DML}} = -0.108$, $R = 3.13$; Scenario B in

3. U.S. CDC NHANES, https://www.cdc.gov/nchs/nhanes/.

93% of resamples). Either marginal defendants, whose sentence depends on the judge's propensity, benefit more from EM than the average defendant, or DML misses a confounder; by Proposition 4(c), the data cannot say which. Its $r^* = 0.244$ lies near the upper edge of the canonical domain $(0.15, 0.25)$, which is why it passes narrowly there ($\widehat{p}_B = 0.958$) and clearly under $(0, 0.8)$ (0.998). `percJudgeSentToEM`, a broader judge-rate measure, brings IV and DML into agreement (-0.138 and -0.101, $R = 1.37$; Scenario A), but only narrowly: 68% of its resamples land in Scenario B.

6. Discussion

DIAL rests on two maintained assumptions: the domain is correctly specified ($\rho_{DU} \in \mathcal{D}$), and the triangular model is linear. It does not need DML to be unconfounded, because residual confounding sends a valid instrument to Scenario B, not to C. We recommend reporting $\hat{\mathcal{C}}_{\text{MCUB}}$, $\widehat{p}_B$, and r^* alongside the binary verdict and flagging instruments in the grey region $0 < \delta < \delta^*$ as "borderline". Scenario A means that IV and DML can be used interchangeably. Scenario B means that the analyst must choose the target: a complier effect from IV, or a population effect from DML if its covariates are trusted. Scenario C means falling back to $\hat{\beta}_{\text{DML}}$ or seeking a better instrument.

Practical guidance. Sign elicitation (Section 4.4) is the single most consequential researcher input: projecting over the wrong half of $(-1, 1)$ can collapse coverage even for a valid instrument (Corollary 3(c)). Because the verdict reduces to whether r^* lies in $\mathcal{D}$, reporting r^* with its interval shows how much the verdict rests on the domain. In the Di Tella showcase, DIAL points the analyst to `percJudgeSentToEM` for population inference, with the caveat that its ratio is imprecise, and flags the estimate from `judgeAlreadyUsedEM` as local or confounded. In deployed ML systems, where instruments are often inherited rather than designed, this separation shows which estimator to trust before acting on a causal estimate.

Limitations. The linearity assumption comes from the CR test's identification of ρ_{ZU}; extending the validity coordinate to nonparametric triangular models is an open problem. With one instrument, DIAL cannot separate a local effect from residual DML confounding (Proposition 4(c)); that needs a second instrument or a covariate that shifts compliance. The binary verdict can hide the continuous severity of an exclusion violation; reporting $\hat{\mathcal{C}}_{\text{MCUB}}$ and the signed distance $d(0, \hat{\mathcal{C}}_{\text{MCUB}})$, as in Figure 4, reduces this risk. The ratio gate has no bootstrap rule, so its A and B labels are much less certain than the validity verdicts, and its threshold $\tau_R = 1.5$ is a convention. Finally, our evidence on ML systems is thin: MovieLens is the only recommender dataset, and we have not tested DIAL inside a deployed recommender or an off-policy evaluation pipeline.

Extensions and future work. A natural extension pairs DIAL with complier-weighted CATE estimation (Oprescu & Kallus, 2024): DIAL would screen validity first, and Scenario B would trigger the compliance correction. A bootstrap rule for the ratio gate, like the one for validity, could withhold the A or B label when neither is stable. Extending the CR test to partially linear or nonparametric triangular models would broaden the validity coordinate. DIAL could also serve as a pre-deployment gate in automated ML pipelines, flagging Scenario C instruments before anyone acts on IV-based estimates.

References

Augustin, T., Coolen, F.P.A., de Cooman, G., & Troffaes, M.C.M. (Eds.). (2014). *Introduction to Imprecise Probabilities*. Wiley.

Banerjee, A. & Iyer, L. (2005). History, institutions, and economic performance: The legacy of colonial land tenure systems in India. *American Economic Review*, 95(4):1190–1213.

Bei, X. (2024). Inference on Union Bounds. *SSRN Working Paper*. Available at SSRN: https://ssrn.com/abstract=5023605 or http://dx.doi.org/10.2139/ssrn.5023605.

Bugni, F.A. (2010). Bootstrap inference in partially identified models defined by moment inequalities: Coverage of the identified set. *Econometrica*, 78(2):735–753.

Burde, D. & Linden, L.L. (2013). Bringing education to Afghan girls: A randomized controlled trial of village-based schools. *American Economic Journal: Applied Economics*, 5(3):27–40.

Card, D. (1995). Using geographic variation in college proximity to estimate the return to schooling. In L.N. Christofides, E.K. Grant, & R. Swidinsky (Eds.), *Aspects of Labour Market Behaviour: Essays in Honour of John Vanderkamp*, pp. 201–222. University of Toronto Press.

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., & Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *Econometrics Journal*, 21(1):C1–C68.

Cinelli, C. & Hazlett, C. (2020). Making sense of sensitivity: Extending omitted variable bias. *Journal of the Royal Statistical Society Series B*, 82(1):39–67.

Conley, T.G., Hansen, C.B., & Rossi, P.E. (2012). Plausibly exogenous. *Review of Economics and Statistics*, 94(1):260–272.

Di Tella, R. & Schargrodsky, E. (2013). Criminal recidivism after prison and electronic monitoring. *Journal of Political Economy*, 121(1):28–73.

DiTraglia, F.J. & García-Jimeno, C. (2021). A framework for eliciting, incorporating, and disciplining identification beliefs in linear models. *Journal of Business & Economic Statistics*, 39(4):1038–1053.

Dzhumashev, R. & Tursunalieva, A. (2026). A test for instrumental variable validity using a partially identified correlation restriction. *Econometric Reviews*, forthcoming. DOI: https://doi.org/10.1080/07474938.2026.2741548.

Foster, D.J. & Syrgkanis, V. (2023). Orthogonal statistical learning. *Annals of Statistics*, 51(3):879–908.

Galiani, S., Rossi, M.A., & Schargrodsky, E. (2011). Conscription and crime: Evidence from the Argentine draft lottery. *American Economic Journal: Applied Economics*, 3(2):119–136.

Gretton, A., Fukumizu, K., Teo, C.H., Song, L., Schölkopf, B., & Smola, A.J. (2008). A kernel statistical test of independence. In *Advances in Neural Information Processing Systems*, vol. 20.

Gunsilius, F.F. (2021). Nontestability of instrument validity under continuous treatments. *Biometrika*, 108(4):989–995.

Hansen, L.P. (1982). Large sample properties of generalized method of moments estimators. *Econometrica*, 50(4):1029–1054.

Harper, F.M. & Konstan, J.A. (2015). The MovieLens Datasets: History and context. *ACM Transactions on Interactive Intelligent Systems*, 5(4):1–19.

Heckman, J.J. & Vytlacil, E. (2005). Structural equations, treatment effects, and econometric policy evaluation. *Econometrica*, 73(3):669–738.

Imbens, G.W. & Angrist, J.D. (1994). Identification and estimation of local average treatment effects. *Econometrica*, 62(2):467–475.

Imbens, G.W. & Manski, C.F. (2004). Confidence intervals for partially identified parameters. *Econometrica*, 72(6):1845–1857.

Jones, M.C. & Pewsey, A. (2026). The eschewed sinh-arcsinh t distribution. *Statistics & Probability Letters*, 228:110560.

Kennedy, E.H. (2022). Semiparametric doubly robust targeted double machine learning: A review. *arXiv:2203.06469*.

Künzel, S.R., Sekhon, J.S., Bickel, P.J., & Yu, B. (2019). Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the National Academy of Sciences*, 116(10):4156–4165.

Manski, C.F. (2003). *Partial Identification of Probability Distributions*. Springer, New York.

Oprescu, M. & Kallus, N. (2024). Estimating heterogeneous treatment effects by combining weak instruments and observational data. *Advances in Neural Information Processing Systems*, 37 (NeurIPS 2024). arXiv:2406.06452.

Papke, L.E. (1995). Participation in and contributions to 401(k) pension plans. *Journal of Human Resources*, 30(2):311–325.

Pearl, J. (1995). On the testability of causal models with latent and instrumental variables. In *Proceedings of the 11th Conference on Uncertainty in Artificial Intelligence*, pp. 435–443.

Romano, J.P., Shaikh, A.M., & Wolf, M. (2014). A practical two-step method for testing moment inequalities. *Econometrica*, 82(5):1979–2002.

Rosenbaum, P.R. (2014). Weighted M-statistics with superior design sensitivity in matched observational studies with multiple controls. *Journal of the American Statistical Association*, 109(507):1145–1158.

Sargan, J.D. (1958). The estimation of economic relationships using instrumental variables. *Econometrica*, 26(3):393–415.

Schölkopf, B., Locatello, F., Bauer, S., Ke, N.R., Kalchbrenner, N., Goyal, A., & Bengio, Y. (2021). Toward causal representation learning. *Proceedings of the IEEE*, 109(5):612–634.

Stoye, J. (2009). More on confidence intervals for partially identified parameters. *Econometrica*, 77(4):1299–1315.

Tamer, E. (2010). Partial identification in econometrics. *Annual Review of Economics*, 2:167–195.

van der Vaart, A. W. and J. A. Wellner (1996). *Weak Convergence and Empirical Processes: With Applications to Statistics*. Springer, New York.

Wager, S. & Athey, S. (2018). Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523):1228–1242.

Walley, P. (1991). *Statistical Reasoning with Imprecise Probabilities*. Chapman & Hall.

Xie, F., Cai, R., Huang, B., Glymour, C., Hao, Z., & Zhang, K. (2020). Generalized independent noise condition for estimating latent variable causal graphs. In *Advances in Neural Information Processing Systems*, vol. 33.

Xie, F., He, Y., Geng, Z., Chen, Z., Hou, R., & Zhang, K. (2022). Testability of instrumental variables in linear non-Gaussian acyclic causal models. *Entropy*, 24(4):512.

Yadlowsky, S., Namkoong, H., Basu, S., Duchi, J., & Tian, L. (2022). Bounds on the conditional and average treatment effect with unobserved confounding factors. *Annals of Statistics*, 50(5):2587–2615.

Zhang, K., Peters, J., Janzing, D., & Schölkopf, B. (2011). Kernel-based conditional independence test and application in causal discovery. In *Proceedings of the 27th Conference on Uncertainty in Artificial Intelligence*.

Zhao, Q., Small, D.S., & Rosenbaum, P.R. (2018). Cross-screening in observational studies that test many hypotheses. *Journal of the American Statistical Association*, 113(523):1070–1084.

Appendix A. Notation

Table 2 summarises the symbols used throughout the paper. It groups them into observed variables; estimators; latent variables and correlation parameters; the CR mapping and identified sets; and DIAL diagnostic quantities and inference parameters.

Table 2: Symbols used in the paper.

Symbol	Meaning	Symbol	Meaning
Observed variables		*Estimators*	
Y	outcome variable	$\hat{\beta}_{\text{IV}}$	IV (2SLS) estimate of the treatment effect
D	treatment (endogenous regressor)	$\hat{\beta}_{\text{DML}}$	DML estimate of the treatment effect
Z	candidate instrument	β_{LATE}	local average treatment effect (IV estimand)
X	observed covariates (controls)	θ_{DML}	DML estimand; β_{ATE} under homogeneous effects
n	sample size	β_{ATE}	average treatment effect (population estimand)
F	first-stage F-statistic (relevance gate)	β_i	individual (heterogeneous) treatment effect
Latent variables & correlation parameters		*CR mapping & identified sets*	
U	structural error; $Y - \beta_{\text{LATE}} D$ net of X	$\boldsymbol{\rho}$	$(\rho_{DY}, \rho_{DZ}, \rho_{YZ})'$, observable correlations
ε	first-stage error ($D = \pi Z + \varepsilon$)	$g(\boldsymbol{\rho}, r)$	CR mapping (Equation 1)
ρ_{ZU}	instrument–error correlation (validity parameter)	$\mathcal{C}$	plug-in identified set $\{g(\boldsymbol{\rho}, r) : r \in \mathcal{D}\}$
ρ_{DU}	treatment–error correlation (endogeneity)	$\hat{\mathcal{C}}_{\text{MCUB}}$	MCUB confidence set for ρ_{ZU}
δ	exclusion-violation magnitude $\lvert\rho_{ZU}\rvert$	V	validity verdict $\mathbf{1}\{\widehat{p}_B \geq \tau_V\}$
σ	std. deviation of the correlation estimator; argument of $\delta^*(n, B, \sigma, \pi^*)$	$\mathcal{D}$	maintained domain for ρ_{DU}
DIAL *diagnostic quantities and inference parameters*			
τ_R	ratio threshold separating Scenarios A and B (default 1.5)	B	number of bootstrap resamples in the validity rule (500)
τ_V	bootstrap coverage threshold for the validity gate (0.95)	π^*	target error level for classification bounds (type-I and type-II)
α_0	MCUB nominal significance level	δ^*	classifiability frontier $\delta^*(n, B, \sigma, \pi^*)$: minimum detectable δ
κ	distance $d(0, C(\rho, \overline{\mathcal{D}}))$ from zero to the identified set	δ^*_∞	population detection threshold $\inf\{\delta : \kappa > 0\}$
κ^*_N	closed-form frontier on the κ scale (Remark 5)	$\widehat{p}_B$	share of the B bootstrap resamples whose MCUB set contains zero
R	IV/DML ratio $\lvert\hat{\beta}_{\text{IV}}/\hat{\beta}_{\text{DML}}\rvert$	r^*	implied endogeneity, the zero of $g(\boldsymbol{\rho}, \cdot)$ (Proposition 4)

Appendix B. Full Empirical Results

Table 3: DIAL applied to 13 instruments across 8 published studies, each under its sign-elicited canonical domain $\mathcal{D}$ (Section 4.4). Five instruments pass the validity gate, and eight fail it; no verdict changes under within-sign widening of $\mathcal{D}$ (Table 4). F is the first-stage F-statistic from Stage 1 of Algorithm 1. Two of the eight invalid instruments also fail this relevance gate, so DIAL labels them C (weak). Bold F values fall below 10, and bold R values lie outside $(1/\tau_R, \tau_R)$, with $\tau_R = 1.5$. $\widehat{p}_B$ is the share of $B = 500$ bootstrap resamples whose MCUB set contains zero; an instrument passes the validity gate when $\widehat{p}_B \geq \tau_V = 0.95$. We analyse Galiani's `highnumber` and Banerjee's `instru` under sign-elicited negative canonical domains; projecting over $(0, 0.8)$ instead would give spuriously invalid verdicts, the failure mode formalised in Corollary 3(c). The NHANES rows use the frame built by `01_nhanes_build.R`, and the Card row uses the 2,320 observations with father's education recorded. Table 5 gives bootstrap intervals for R, Appendix G gives replication details, and Appendix B.3 discusses each verdict.

Study	Instrument	n	$\mathcal{D}$	F	R	$\hat{\mathcal{C}}_{\text{MCUB}}$	$\widehat{p}_B$	DIAL
Di Tella & Schargrodsky (2013): Criminal Recidivism								
	`judgeAlreadyUsedEM`	1,503	$(0.15, 0.25)$	19.0	**3.13**	$[-0.544, 0.454]$	0.958	**B**
	`percJudgeSentToEM`	1,503	$(0, 0.25)$	98.9	1.37	$[-0.204, 0.365]$	0.996	**A**
MovieLens 100K: Recommendation Effects[†]								
	`num_genres`[†]	1,682	$(0, 0.8)$	163.1	**0.62**[¶]	$[-0.400, 0.695]$	1.000	**B** (sign)
	`release_year`[†‡]	1,681	$(-0.8, 0)$	**1.7**	—	$[-1.000, -0.244]$	0.016	**C** (weak)
	`is_sequel`[†‡]	1,682	$(-0.8, 0)$	**1.3**	—	$[-1.000, -0.524]$	0.000	**C** (weak)
Galiani et al. (2011): Conscription and Crime								
	`highnumber`	5,000	$(-0.2, 0)$	67,121	**2.44**	$[-0.194, 0.016]$	1.000	**B**
Papke (1995): 401(k) Participation								
	`sole_plan`	1,534	$(0, 0.3)$	22.2	—	$[0.101, 0.964]$	0.284	**C**
	`plan_age`	1,534	$(0, 0.3)$	27.8	—	$[0.663, 1.000]$[§]	0.000	**C**
NHANES Fish Consumption (Zhao et al. 2018 extract)								
	`income`	1,107	$(0, 0.8)$	61.0	—	$[0.511, 1.000]$[§]	0.000	**C**
	`education`	1,107	$(0, 0.8)$	92.2	—	$[0.157, 1.000]$[§]	0.062	**C**
Card Returns to Education								
	`fatheduc`	2,320	$(0, 0.4)$	204.6	—	$[-0.071, 0.584]$[♯]	0.814	**C**
Burde & Linden (2013): School Construction								
	`buildschool`	687	$(0, 0.4)$	257.1	—	$[0.146, 0.736]$	0.006	**C**
Banerjee & Iyer (2005): Colonial Land Tenure								
	`instru`	3,541	$(-0.6, -0.4)$	360.1	**1.84**	$[-0.162, 0.224]$	1.000	**B**

[†] The MovieLens analysis aggregates the 100,000 ratings to the item level (1,682 movies; 1,681 for `release_year`, which is missing for one movie), because `num_genres`, `release_year`, and `is_sequel` are item-level features. Each movie contributes its mean rating, its log number of ratings, and, as a control, the mean leniency of its raters. The CR test works on the cross-section of (instrument, treatment, outcome) at the unit of variation, so the movie is the appropriate unit. At this level, `release_year` and `is_sequel` fail the relevance gate (first-stage $F = 1.7$ and 1.3), so Algorithm 1 returns Scenario C (weak); their MCUB sets also exclude zero.

[‡] MCUB lower endpoints of -1.042 (`release_year`) and -1.072 (`is_sequel`) are clipped to -1.000 because $\rho_{ZU} \in [-1, 1]$; the verdicts are unaffected.

[§] MCUB upper endpoints of 1.021 (Papke `plan_age`), 1.053 (NHANES `income`), and 1.015 (NHANES `education`) are clipped to 1.000; the verdicts are unaffected.

[♯] The MCUB set for Card `fatheduc` contains zero, but only 81% of the resampled sets do, so $\widehat{p}_B < \tau_V$ and DIAL returns Scenario C. Card is the one instrument on which the bootstrap rule and the single-sample check disagree.

[¶] The 2SLS and DML estimates for `num_genres` have opposite signs (-0.038, SE 0.039; 0.061, SE 0.017), so Algorithm 1 returns Scenario B (sign); $R = 0.62 < 1/\tau_R$ would also give B. The 2SLS interval covers zero, so the disagreement may reflect an imprecise IV estimate.

B.1. Domain sensitivity

Table 4: Domain sensitivity under the within-sign protocol (Section 4.4, step 3). Each instrument's canonical $\mathcal{D}$ is paired with a robustness $\mathcal{D}$ inside the same structurally admissible half-line. The three MovieLens rows and the two NHANES rows have no robustness pair, because their canonical $\mathcal{D}$ is already the widest admissible subinterval of the elicited half-line. Five instruments are robustly Valid, and eight are robustly Invalid. No verdict changes when $\mathcal{D}$ is widened within its sign. The bootstrap share $\widehat{p}_B$ ($B = 500$) is at least 0.958 for every Valid set and at most 0.814 for every Invalid set. Card's sets contain zero, but the rule rejects them ($\widehat{p}_B = 0.814$ and 0.798).

Instrument	Canonical $\mathcal{D}$	$\hat{\mathcal{C}}_{\text{MCUB}}$ (canonical)	Robustness $\mathcal{D}$	$\hat{\mathcal{C}}_{\text{MCUB}}$ (robust)	Robust?
Di Tella `judgeAlreadyUsedEM`	$(0.15, 0.25)$	$[-0.544,\ 0.454]$ Valid	$(0, 0.8)$	$[-0.614,\ 0.928]$ Valid	Yes
Di Tella `percJudgeSentToEM`	$(0, 0.25)$	$[-0.204,\ 0.365]$ Valid	$(0, 0.8)$	$[-0.202,\ 0.873]$ Valid	Yes
Galiani `highnumber`	$(-0.2, 0)$	$[-0.194,\ 0.016]$ Valid	$(-0.8, 0)$	$[-0.795,\ 0.016]$ Valid	Yes
Banerjee `instru`	$(-0.6, -0.4)$	$[-0.162,\ 0.224]$ Valid	$(-0.8, 0)$	$[-0.433,\ 0.596]$ Valid	Yes
Burde `buildschool`	$(0, 0.4)$	$[0.146,\ 0.736]$ Invalid	$(0, 0.8)$	$[0.146,\ 0.982]$ Invalid	Yes
Card `fatheduc`	$(0, 0.4)$	$[-0.071,\ 0.584]$ Invalid$^\sharp$	$(0, 0.8)$	$[-0.051,\ 0.904]$ Invalid$^\sharp$	Yes
Papke `sole_plan`	$(0, 0.3)$	$[0.101,\ 0.964]$ Invalid	$(0, 0.8)$	$[0.101,\ 1.000]^\ddagger$ Invalid	Yes
Papke `plan_age`	$(0, 0.3)$	$[0.663,\ 1.000]^\ddagger$ Invalid	$(0, 0.8)$	$[0.685,\ 1.000]^\ddagger$ Invalid	Yes
MovieLens `num_genres`	$(0, 0.8)$	$[-0.400,\ 0.695]$ Valid	*widest*$^\oplus$	—	Yes
MovieLens `release_year`	$(-0.8, 0)$	$[-1.000,\ -0.244]^\ddagger$ Invalid	*widest*$^\ominus$	—	Yes
MovieLens `is_sequel`	$(-0.8, 0)$	$[-1.000,\ -0.524]^\ddagger$ Invalid	*widest*$^\ominus$	—	Yes
NHANES `income`	$(0, 0.8)$	$[0.511,\ 1.000]^\ddagger$ Invalid	*widest*$^\oplus$	—	Yes
NHANES `education`	$(0, 0.8)$	$[0.157,\ 1.000]^\ddagger$ Invalid	*widest*$^\oplus$	—	Yes

Note on the MovieLens and NHANES rows. Their canonical domains are already the widest structurally admissible subintervals within their elicited signs. The domain is $(0, 0.8)$ for `num_genres`, with positive endogeneity: genre coverage proxies production investment, which correlates positively with unobserved quality. It is $(-0.8, 0)$ for `release_year` and `is_sequel`, with negative endogeneity: older titles and sequels carry accumulated brand capital that depresses the marginal rating effect. It is $(0, 0.8)$ for NHANES `income` and `education`, with positive endogeneity: both track unobserved healthy-eating habits that also affect mercury exposure. Widening further within the half would admit correlations of magnitude ≥ 0.8, which the structural story rules out. We therefore report these verdicts as *widest* within the admissible half, and no within-sign robustness pair is possible. The $\oplus$/$\ominus$ superscripts denote the elicited sign.

‡ Seven MCUB endpoints fall outside $[-1, 1]$ and are clipped because $\rho_{ZU} \in [-1, 1]$: -1.042 (MovieLens `release_year`), -1.072 (`is_sequel`), 1.082 (Papke `sole_plan`, robust), 1.021 and 1.023 (Papke `plan_age`), 1.053 (NHANES `income`), and 1.015 (NHANES `education`). The verdicts are unaffected.

$^\sharp$ Card's sets contain zero, but only 81% and 80% of the resampled sets do, so the bootstrap rule rejects the instrument in both domains.

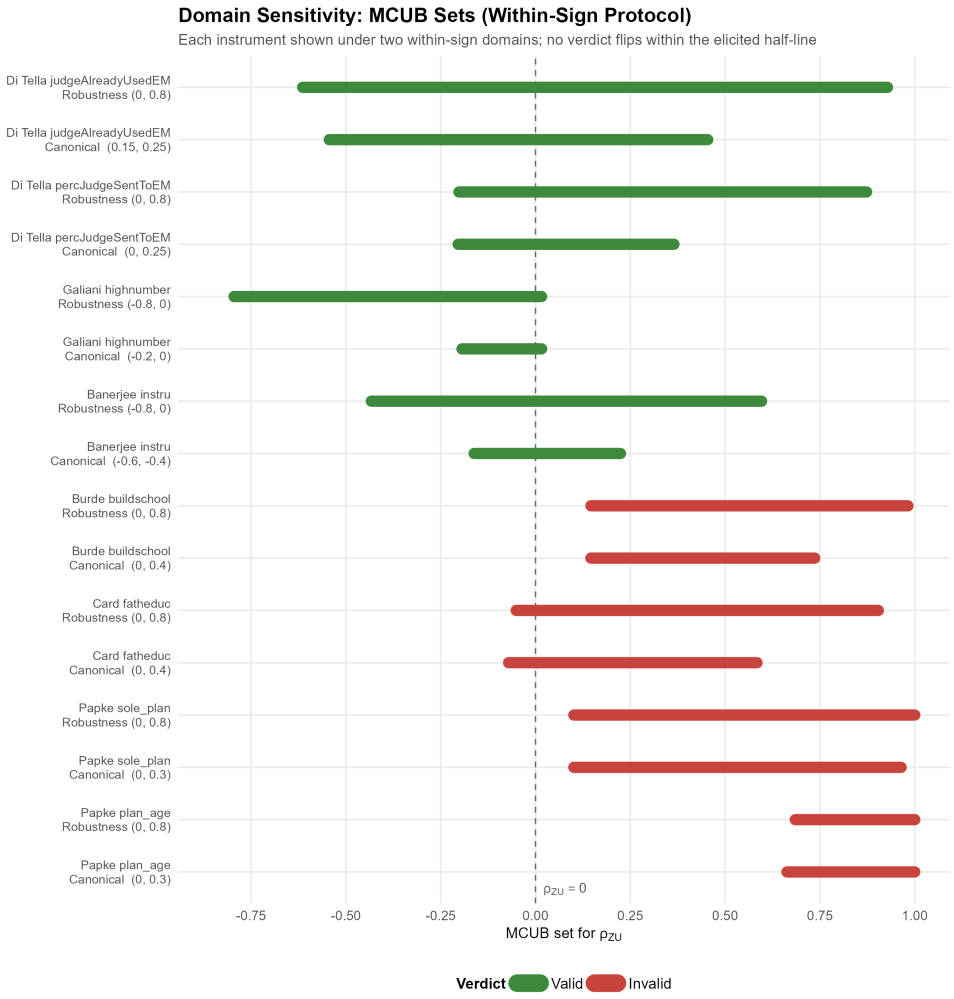


Figure 3: MCUB sets under the within-sign robustness protocol (Section 4.4, step 3). Each of the eight instruments with a robustness pair appears as a pair of bars, one for the canonical $\mathcal{D}$ and one for the robustness $\mathcal{D}$, both within the same structurally admissible half-line. The MovieLens and NHANES instruments have no pair, because their canonical $\mathcal{D}$ is already the widest admissible subinterval. Green indicates a Valid verdict ($\widehat{p}_B \geq \tau_V$), and red an Invalid one. Card's sets contain zero but fail the bootstrap rule. Each pair shares one colour, which confirms that no verdict changes under within-sign widening. Table 4 reports the exact values.

B.2. Implied endogeneity and the IV/DML ratio

Table 5 reports the implied endogeneity r^* of Proposition 4 for all 13 instruments and, for the five that pass the validity gate, the ratio R with a bootstrap interval. In every case, the instrument passes the gate exactly when r^* lies in its domain, the pattern that part (a) of the proposition predicts for the plug-in set. Card's interval for r^*, $[-0.214, 0.073]$, reaches into its domain, which is why 81% of its

resampled MCUB sets still contain zero. The two weak MovieLens instruments have $|r^*|$ close to 1 and intervals that cover almost all of $(-1, 1)$: with a weak first stage, the data fit a valid instrument only under near-perfect endogeneity and barely pin r^* down. The intervals for R are wide. Only `num_genres` and `judgeAlreadyUsedEM` land in Scenario B in more than 90% of resamples, and the A label of `percJudgeSentToEM` holds in only about a third of them.

Table 5: Implied endogeneity r^* (Proposition 4) and the IV/DML ratio R, each instrument under its canonical domain $\mathcal{D}$. Intervals are 95% percentile intervals from bootstrap resamples of the raw rows: 2,000 for r^*, and 200 for R, with DML refitted on each resample. "B share" is the share of these 200 resamples that Algorithm 1 would place in Scenario B. For a valid instrument, r^* measures the gap between OLS and IV in correlation units (Proposition 4(b)).

Study	Instrument	$\mathcal{D}$	r^*	95% interval	R	95% interval	B share	DIAL
Di Tella	`judgeAlreadyUsedEM`	(0.15, 0.25)	+0.244	[−0.205, +0.628]	3.13	[0.32, 11.98]	0.930	B
	`percJudgeSentToEM`	(0, 0.25)	+0.044	[−0.137, +0.221]	1.37	[0.08, 3.94]	0.675	A
MovieLens	`num_genres`	(0, 0.8)	+0.284	[+0.158, +0.420]	0.62	[0.05, 7.79]	0.995	B (sign)
	`release_year`	(−0.8, 0)	−0.987	[−1.000, +1.000]	—	—	—	C (weak)
	`is_sequel`	(−0.8, 0)	+0.983	[−1.000, +1.000]	—	—	—	C (weak)
Galiani	`highnumber`	(−0.2, 0)	−0.011	[−0.019, −0.003]	2.44	[0.93, 19.56]	0.775	B
Papke	`sole_plan`	(0, 0.3)	−0.439	[−0.749, −0.043]	—	—	—	C
	`plan_age`	(0, 0.3)	−0.799	[−0.916, −0.651]	—	—	—	C
NHANES	`income`	(0, 0.8)	−0.694	[−0.825, −0.530]	—	—	—	C
	`education`	(0, 0.8)	−0.339	[−0.504, −0.143]	—	—	—	C
Card	`fatheduc`	(0, 0.4)	−0.067	[−0.214, +0.073]	—	—	—	C
Burde	`buildschool`	(0, 0.4)	−0.272	[−0.399, −0.150]	—	—	—	C
Banerjee	`instru`	(−0.6, −0.4)	−0.530	[−0.602, −0.452]	1.84	[1.05, 6.65]	0.760	B

B.3. Substantive notes on instrument verdicts

401(k) participation (Scenario C). Both instruments, sole plan and plan age, plausibly affect savings outcomes through channels other than match-rate exposure; for example, both may correlate with how attentive the employer is to retirement benefits more broadly. Under the sign-elicited canonical domain $(0, 0.3)$, the MCUB sets are $[0.101, 0.964]$ for sole plan and $[0.663, 1.000]$ for plan age. Both exclude zero, consistent with this prior, and the bootstrap rule rejects both ($\widehat{p}_B = 0.284$ and 0.000). Widening to the robustness domain $(0, 0.8)$ leaves both verdicts unchanged: the sets become $[0.101, 1.000]$ and $[0.685, 1.000]$, so the lower endpoints stay well above zero (Table 4).

NHANES nutritional epidemiology (Scenario C). Income and education affect mercury exposure through diet channels other than fish consumption, including fruit and vegetable intake and supplements. Under the sign-elicited canonical domain $(0, 0.8)$, the widest defensible positive subinterval, the MCUB sets are $[0.511, 1.000]$ for income and $[0.157, 1.000]$ for education (upper endpoints 1.053 and 1.015 before clipping). Both lie well above zero, consistent with positive endogeneity through these channels. Because the canonical domain is already the widest defensible positive domain, no within-sign widening is possible (Table 4).

Card returns to education (Scenario C). The father's-education instrument is robustly Invalid. Under the canonical domain $(0, 0.4)$, the MCUB set is $[-0.071, 0.584]$; under the robustness domain $(0, 0.8)$, it is $[-0.051, 0.904]$ (Table 4). Both sets contain zero, but only 81% and 80% of the resampled sets do, so the bootstrap rule rejects the instrument in both domains. The plug-in identified set, $[0.067, 0.461]$, excludes zero because the implied endogeneity, $r^* = -0.067$, lies just below the

positive domain: a valid instrument would need a slightly negative ability bias. Parental-education networks that act through channels other than schooling-induced income, a long-standing concern in the returns-to-schooling literature, would explain such a violation. Card is the instrument nearest the detection boundary, and the one for which a single MCUB set and the bootstrap rule disagree.

Burde & Linden school construction (Scenario C). The build-school instrument is robustly Invalid. Under the canonical domain $(0, 0.4)$, the MCUB set is $[0.146, 0.736]$; under the robustness domain $(0, 0.8)$, it expands to $[0.146, 0.982]$ (Table 4). The lower endpoint does not move between the two domains; only the upper endpoint follows $\mathcal{D}$. School placement was randomised, so the instrument is independent of village characteristics, and the verdict concerns exclusion relative to enrolment. A village school can raise test scores through channels other than enrolment, for example through shorter travel and more instruction for children who would have enrolled in a distant school anyway. In the data, the instrument would be valid only if enrolment were negatively correlated with the test-score error ($r^* = -0.272$), which the positive domain rules out.

Banerjee colonial land tenure (Scenario B). The data are Indian district-years from 1965 onwards. The treatment is the share of a district's land under non-landlord revenue systems, the outcome is the share of wheat area under high-yielding varieties, and the instrument `instru` (Banerjee & Iyer, 2005) marks districts that came under British revenue administration between 1820 and 1856, when new areas mostly received non-landlord systems. DIAL classifies `instru` as Scenario B under the sign-elicited canonical domain $(-0.6, -0.4)$: it passes the validity gate, but the IV estimate is 1.84 times the DML estimate. Its bootstrap interval for R, $[1.05, 6.65]$, lies above one, although only 76% of resamples exceed τ_R. The instrument remains Valid under the robustness domain $(-0.8, 0)$ (Table 4). The negative domain follows the selection that the original authors suggest: landlord systems were more likely in more fertile areas, so the non-landlord share correlates negatively with unobserved productivity. The implied endogeneity, $r^* = -0.530$, is large and lies inside both domains. The verdict therefore depends on a structural story that admits strong negative endogeneity: a domain that stopped at -0.5 would leave r^* outside it, and the plug-in identified set would exclude zero (Proposition 4(a)). Projecting over a positive domain would misspecify the sign and collapse coverage, the failure mode formalised in Corollary 3(c).

Galiani conscription and crime (Scenario B). The data aggregate to cells of five birth cohorts by 1,000 lottery numbers ($n = 5{,}000$). Draft eligibility moves the share who served sharply across cells, which explains the first-stage $F = 67{,}121$. Both the IV estimate (0.004) and the DML estimate are close to zero, so $R = 2.44$ is a ratio of two small numbers, and its bootstrap interval, $[0.93, 19.56]$, is wide (Table 5). The implied endogeneity, $r^* = -0.011$, lies inside the canonical domain $(-0.2, 0)$, so the instrument passes the validity gate. The Scenario B verdict says that IV and DML disagree in relative terms, not that either effect is large.

MovieLens `release_year` and `is_sequel` (Scenario C). At the item level, both instruments are weak: their first-stage F statistics are 1.7 and 1.3, so Algorithm 1 stops at the relevance gate. Under the sign-elicited canonical domain $(-0.8, 0)$, their MCUB sets, $[-1.000, -0.244]$ and $[-1.000, -0.524]$ (lower endpoints -1.042 and -1.072 before clipping), also exclude zero. This is consistent with a structural exclusion violation: production effort and franchise brand capital plausibly affect ratings through channels other than exposure. Sign elicitation rescues neither instrument.

Appendix C. Proofs

C.1. Proof of Theorem 1 (Separation Theorem)

Proof sketch Part 1: the sets $\{A_n, B_n, C_n\}$ are disjoint and exhaustive by construction, since $B_n = \{V_n = 1\} \setminus A_n$. Part 2 follows directly from Assumption 3: since $A_n \cup B_n = \{V_n = 1\}$, the two inequalities are exactly the two finite-sample statements in Assumption 3. Part 3 combines Assumption 4, which keeps $|\log R_0|$ at least η away from $\log \tau_R$ when $s_0 = 1$, with $R_n \to_p R_0$ and $S_n \to_p s_0$ from Assumption 2, and then applies the union bound with the validity bound from Part 2. ■

If $p_n^\star = m_n(\delta)$ with m_n strictly decreasing, as in Proposition 2(c), then $\Pr_\delta(V_n = 0) = \Pr\big(\mathrm{Bin}(B, m_n(\delta)) < \lceil B\tau_V \rceil\big)$ increases strictly in δ, so the frontier δ^* in Assumption 3 is well defined.

Part 1 (Partition). By construction, $A_n = \{V_n = 1,\ S_n = 1,\ |\log R_n| < \log \tau_R\}$, $B_n = \{V_n = 1\} \setminus A_n$, and $C_n = \{V_n = 0\}$. A_n and B_n are disjoint by definition, and $A_n \cap C_n = B_n \cap C_n = \varnothing$ because V_n cannot equal both 0 and 1. On $\{F \geq 10\}$, either $V_n = 0$ (giving C_n) or $V_n = 1$, and then the outcome lies in A_n or in its complement within $\{V_n = 1\}$, which is B_n. Hence $A_n \cup B_n \cup C_n$ exhausts the sample space conditional on $\{F \geq 10\}$.

Part 2 (Validity separation, finite-sample). Since $A_n \cup B_n = \{V_n = 1\}$ and $C_n = \{V_n = 0\}$, Assumption 3 gives directly

$$\Pr_{\delta=0}(A_n \cup B_n) = \Pr_{\delta=0}(V_n = 1) \geq 1 - \pi^*, \qquad \Pr_{\delta \geq \delta^*}(C_n) = \Pr_{\delta \geq \delta^*}(V_n = 0) \geq 1 - \pi^*.$$

Both bounds hold at every finite n. Part 2 is therefore a direct consequence of Assumption 3. Its non-tautological content is that Assumption 3 actually holds and that the implied δ^* and $\alpha_n(\delta)$ take explicit finite-sample forms. Proposition 2 supplies this content; its Hoeffding bound operationalises the assumption.

Part 3 (Agreement separation, asymptotic). Suppose $\delta = 0$ and the population is in agreement (A*): $s_0 = 1$ and, by Assumption 4, $|\log R_0| \leq \log \tau_R - \eta$. By Assumption 2, $S_n \to_p 1$ and, because $R_0 > 0$, $\log R_n \to_p \log R_0$. Hence $\Pr(S_n = -1 \text{ or } |\log R_n| \geq \log \tau_R \mid \delta = 0, \mathrm{A}^*) = o(1)$. Applying the union bound,

$$\begin{aligned}
&\Pr(A_n^c \mid \delta = 0,\ \mathrm{A}^*) \\
&\quad \leq \Pr(V_n = 0 \mid \delta = 0) + \Pr(S_n = -1 \text{ or } |\log R_n| \geq \log \tau_R \mid \delta = 0,\ \mathrm{A}^*) \\
&\quad \leq \pi^* + o(1),
\end{aligned}$$

applying Assumption 3 to the first term. Hence $\Pr(A_n \mid \delta = 0, \mathrm{A}^*) \geq 1 - \pi^* - o(1)$.

Now suppose the population is in disagreement (B*). If $s_0 = -1$, then $\Pr(S_n = 1) = o(1)$. If $s_0 = 1$, then Assumption 4 gives $|\log R_0| \geq \log \tau_R + \eta$, so $\Pr(|\log R_n| < \log \tau_R) = o(1)$. In both cases, $\Pr(A_n) \leq \Pr(S_n = 1,\ |\log R_n| < \log \tau_R) = o(1)$. Since $B_n^c = C_n \cup A_n$, the union bound gives

$$\begin{aligned}
&\Pr(B_n^c \mid \delta = 0,\ \mathrm{B}^*) \\
&\quad \leq \Pr(V_n = 0 \mid \delta = 0) + \Pr(S_n = 1,\ |\log R_n| < \log \tau_R \mid \delta = 0,\ \mathrm{B}^*) \\
&\quad \leq \pi^* + o(1),
\end{aligned}$$

so $\Pr(B_n \mid \delta = 0, \mathrm{B}^*) \geq 1 - \pi^* - o(1)$. In both cases, the $o(1)$ term comes from the consistency of R_n and S_n and vanishes as $n \to \infty$ for any fixed $\eta > 0$.

Conditioning remarks. All three parts are conditional on $\{F \geq 10\}$ and on $\mathcal{D}$ containing the true ρ_{DU}. If $\mathcal{D}$ is misspecified, the MCUB set need not contain zero even for a valid instrument (Corollary 3), and the bounds in Parts 2 and 3 can fail. □

C.2. Proof of Proposition 2

Proof sketch Conditional on $\mathcal{X}_n$, the variables $I_1, \ldots, I_B$ are i.i.d. Bernoulli$(p_n^\star)$, so Hoeffding's inequality yields $\Pr^\star(E_{MC} \mid \mathcal{X}_n) \leq \exp(-2B|p_n^\star - \tau_V|^2)$. For Part (b), on the invalid side let $A_\epsilon^- := \{p_n^\star \leq \tau_V - \epsilon\}$; then $\Pr_\delta(\text{classify as Valid}) \leq e^{-2B\epsilon^2} + \Pr_\delta((A_\epsilon^-)^c)$, so setting $\epsilon = \epsilon_H(\pi^*) := \sqrt{\log(1/\pi^*)/(2B)}$ gives the bound. On the valid side, the exact binomial tail replaces Hoeffding's bound. Part (c) follows by inverting the strictly decreasing calibration curve at $\tau_V - \epsilon_H(\pi^*)$. ■

Remark 5 (Approximate closed-form frontier) *Let α_0 denote the nominal significance level of the MCUB set, inherited from the CR construction (Dzhumashev & Tursunalieva, 2026). The empirical applications and the power curves use $\alpha_0 = 0.05$, and the other simulations $\alpha_0 = 0.10$; this parameter is separate from the classification-error level π^*. Under partial identification, coverage depends on the violation through the distance $\kappa = d\big(0, C(\rho, \overline{\mathcal{D}})\big)$ of Corollary 3(b). This distance is zero whenever some $r \in \mathcal{D}$ rationalises the data with a valid instrument. Under the Gaussian approximations $\mathrm{CP}_n \approx \Phi(z_{1-\alpha_0/2} - \sqrt{n}\,\kappa/\sigma)$ and $\hat{p} \approx \mathcal{N}(p, p(1-p)/B)$, with a conservative SE evaluated at τ_V, the frontier on the distance scale has the closed form*

$$\kappa_N^*(n, B, \sigma, \pi^*) \approx \frac{\sigma}{\sqrt{n}} \left[z_{1-\alpha_0/2} - \Phi^{-1}\left(\tau_V - z_{1-\pi^*} \sqrt{\frac{\tau_V(1-\tau_V)}{B}} \right) \right]. \tag{2}$$

The frontier on the δ scale is the smallest δ whose identified set lies at least κ_N^* from zero. It therefore exceeds $\delta_\infty^*(\mathcal{D}) := \inf\{\delta : \kappa > 0\}$, which does not shrink with n. The two nominal levels play different roles: α_0 controls the width of the MCUB set in each sample*, while π^* controls the classification error rate of* DIAL *across samples. The formula is approximate, not exact, because it relies on both Gaussian approximations.*

Proof of Proposition 2

Part (a). Suppose $p_n^\star < \tau_V$. Conditional on $\mathcal{X}_n$,

$$E_{FA} = \{\widehat{p}_B - p_n^\star \geq \tau_V - p_n^\star\}.$$

Since $\widehat{p}_B$ is the average of B conditionally i.i.d. Bernoulli$(p_n^\star)$ variables, the one-sided Hoeffding inequality yields

$$\overset{\star}{\Pr}(E_{FA} \mid \mathcal{X}_n) \leq \exp\big(-2B(\tau_V - p_n^\star)^2\big).$$

The case $p_n^\star > \tau_V$ is symmetric: $E_{FR} = \{p_n^\star - \widehat{p}_B > p_n^\star - \tau_V\}$, giving

$$\overset{\star}{\Pr}(E_{FR} \mid \mathcal{X}_n) \leq \exp\big(-2B(p_n^\star - \tau_V)^2\big).$$

Combining both cases gives the stated bound for $p_n^\star \neq \tau_V$.

The exact binomial probabilities follow because $B\widehat{p}_B \sim \mathrm{Bin}(B, p_n^\star)$ conditional on $\mathcal{X}_n$, so $\{\widehat{p}_B \geq \tau_V\} = \{B\widehat{p}_B \geq \lceil B\tau_V \rceil\}$ and $\{\widehat{p}_B < \tau_V\} = \{B\widehat{p}_B \leq \lceil B\tau_V \rceil - 1\}$.

Part (b). Fix $\epsilon > 0$ and let $A_{\epsilon,n} := \{|p_n^\star - \tau_V| \geq \epsilon\}$. By the law of iterated expectations,

$$\Pr_\delta(E_{MC}) = \mathbb{E}_\delta\big[\overset{\star}{\Pr}(E_{MC} \mid \mathcal{X}_n)\mathbf{1}_{A_{\epsilon,n}}\big] + \mathbb{E}_\delta\big[\overset{\star}{\Pr}(E_{MC} \mid \mathcal{X}_n)\mathbf{1}_{A^c_{\epsilon,n}}\big].$$

Part (a) bounds the first term by $e^{-2B\epsilon^2}$, and the second by $\Pr_\delta(A^c_{\epsilon,n})$, giving

$$\Pr_\delta(E_{MC}) \leq e^{-2B\epsilon^2} + \Pr_\delta(A^c_{\epsilon,n}).$$

Setting $\epsilon = \epsilon_H(\pi^*) := \sqrt{\log(1/\pi^*)/(2B)}$ yields $e^{-2B\epsilon_H(\pi^*)^2} = \pi^*$, hence

$$\Pr_\delta(E_{MC}) \leq \pi^* + \Pr_\delta(A^c_{\epsilon_H(\pi^*),n}).$$

On the invalid side, the hypothesis gives $\Pr_\delta(p_n^\star \leq \tau_V - \epsilon_H(\pi^*)) \geq 1 - \alpha_n(\delta)$ for $\delta \geq \delta_H^*$. That event is contained in $A_{\epsilon_H(\pi^*),n}$, so $\Pr_\delta(A^c_{\epsilon_H(\pi^*),n}) \leq \alpha_n(\delta)$, and therefore

$$\Pr_\delta(E_{FA}) \leq \pi^* + \alpha_n(\delta), \qquad \Pr_\delta(\text{classify as Invalid}) \geq 1 - \pi^* - \alpha_n(\delta).$$

On the valid side ($\delta = 0$), the misclassification event is $E_{FR} = \{\widehat{p}_B < \tau_V\}$, and $\Pr^\star(E_{FR} \mid \mathcal{X}_n) = \Pr\big(\mathrm{Bin}(B, p_n^\star) \leq \lceil B\tau_V \rceil - 1\big)$, which decreases in $p_n^\star$. Let $p_{\min}$ solve $\Pr\big(\mathrm{Bin}(B, p_{\min}) \leq \lceil B\tau_V \rceil - 1\big) = \pi^*$. On the event $\{p_n^\star \geq p_{\min}\}$, the conditional error is at most π^*, so the same decomposition gives

$$\Pr_0(E_{FR}) \leq \pi^* + \Pr_0(p_n^\star < p_{\min}) \leq \pi^* + \alpha_n(0)$$

whenever $\Pr_0(p_n^\star \geq p_{\min}) \geq 1 - \alpha_n(0)$. The Hoeffding analogue of this step would need $p_n^\star \geq \tau_V + \epsilon_H(\pi^*)$, which no probability can satisfy when $\tau_V + \epsilon_H(\pi^*) > 1$. This happens at $\tau_V = 0.95$ and $B = 500$ for every $\pi^* < 0.08$. The exact thresholds are $p_{\min} = 0.9685$, 0.9633, and 0.9604 at $\pi^* = 0.01$, 0.05, and 0.10.

Part (c). If $p_n^\star = m_n(\delta)$ a.s. with m_n continuous and strictly decreasing, then $p_n^\star \leq \tau_V - \epsilon_H(\pi^*)$ if and only if $\delta \geq m_n^{-1}(\tau_V - \epsilon_H(\pi^*))$, which identifies $\delta_H^*(n, B, \pi^*) = m_n^{-1}(\tau_V - \epsilon_H(\pi^*))$.

Relation to Theorem 1. Proposition 2 operationalises Assumption 3. Part (b) above verifies that the bootstrap verdict V_n satisfies the classification-error inequalities of Assumption 3, with the specific threshold δ_H^* and slack $\alpha_n(\delta)$ identified here. Part 2 of Theorem 1 is a direct consequence of Assumption 3, and this proposition supplies the assumption's content. Part 2 of Theorem 1 therefore draws its non-tautological force from the Hoeffding bound in Part (a) above. ■

C.3. Proof of Corollary 3

Throughout, $\widehat{C}_n(\mathcal{D})$ abbreviates $\hat{\mathcal{C}}_{\mathrm{MCUB},n}(\mathcal{D})$, and $\widehat{C}_n$ drops the domain when it is fixed. We first establish the Hausdorff-consistency result on which Part (b) relies (Lemma 6) and then prove each part in turn. Part (a) holds because domain monotonicity makes $\{0 \in \hat{\mathcal{C}}_{\mathrm{MCUB},n}(\mathcal{D})\}$ monotone in $\mathcal{D}$. Part (b) uses compactness: if $0 \notin C$, then $\kappa(\mathcal{D}) > 0$, so Hausdorff consistency sends the probability to zero. Part (c) specialises (b) to the unique-zero case.

Lemma 6 (Hausdorff consistency of the MCUB set) *Let $\overline{\mathcal{D}} \subset (-1, 1)$ be compact. Suppose (H1) $\hat{g}_n(\hat{\rho}_n, r) \to_p g(\rho_0, r)$ for each fixed $r \in \overline{\mathcal{D}}$; (H2) $r \mapsto \hat{g}_n(\hat{\rho}_n, r)$ is stochastically equicontinuous on $\overline{\mathcal{D}}$; (H3) $r \mapsto g(\rho_0, r)$ is continuous; and (H4) the MCUB set expands the plug-in image by a vanishing margin: almost surely, $\hat{\mathcal{C}}_{MCUB,n}(\mathcal{D}) = \bigcup_{r\in\overline{\mathcal{D}}} \big[\hat{g}_n(\hat{\rho}_n, r) - \hat{e}_n(r),\ \hat{g}_n(\hat{\rho}_n, r) + \hat{e}_n(r)\big]$ with $\hat{e}_n(r) \geq 0$ and $\bar{e}_n := \sup_{r\in\overline{\mathcal{D}}} \hat{e}_n(r) \to_p 0$. Then $d_H(\hat{\mathcal{C}}_{MCUB,n}(\mathcal{D}),\ C(\rho_0, \overline{\mathcal{D}})) \to_p 0$.*

Proof Write $\hat{g}_n(r) \equiv \hat{g}_n(\hat{\rho}_n, r)$, $g(r) \equiv g(\rho_0, r)$, and $\widehat{C}_n^0 := \{\hat{g}_n(r) : r \in \overline{\mathcal{D}}\}$ for the plug-in image. By (H1)–(H3) and compactness of $\overline{\mathcal{D}}$, the standard Arzelà–Ascoli finite-cover argument yields the uniform convergence $\Delta_n := \sup_{r\in\overline{\mathcal{D}}} |\hat{g}_n(r) - g(r)| \to_p 0$. Matching $\hat{g}_n(r)$ with $g(r)$ for each r bounds both directed distances between $\widehat{C}_n^0$ and C by Δ_n, so $d_H(\widehat{C}_n^0, C) \leq \Delta_n$. By (H4), $\widehat{C}_n^0 \subseteq \widehat{C}_n$, and every point of $\widehat{C}_n$ lies within $\bar{e}_n$ of $\widehat{C}_n^0$, so $d_H(\widehat{C}_n, \widehat{C}_n^0) \leq \bar{e}_n$. The triangle inequality gives $d_H(\widehat{C}_n, C) \leq \Delta_n + \bar{e}_n \to_p 0$. ■

Remark 7 (Sufficient conditions for (H1), (H2), and (H4)) *(H1) holds whenever $\hat{\rho}_n \to_p \rho_0$ and g is continuous in ρ. (H2) holds when $r \mapsto \hat{g}_n(\hat{\rho}_n, r)$ is a sample average of Lipschitz-in-r functions with square-integrable Lipschitz constant (van der Vaart and Wellner, 1996, Lemma 2.14.1), or when $\{\hat{g}_n(\cdot, r) : r \in \overline{\mathcal{D}}\}$ is a Donsker class. (H4) holds for the MCUB set with $\hat{e}_n(r) = \hat{c}_n\, \widehat{\mathrm{se}}_n(r)$ when the critical value $\hat{c}_n$ is $O_p(1)$ and $\sup_r \widehat{\mathrm{se}}_n(r) = O_p(n^{-1/2})$.*

Remark 8 (Maintained domain as a credal / robust-Bayes prior) *In our formulation, the maintained domain $\mathcal{D}$ is a frequentist sensitivity parameter, but it maps naturally onto a prior over the unobserved endogeneity ρ_{DU}. Two contrasts clarify the probabilistic structure. (i) A Bayesian credible-set approach would place a proper prior on ρ_{DU} supported on $\mathcal{D}$, integrate out ρ_{DU} to obtain a posterior on ρ_{ZU}, and report a Bayesian credible interval; its coverage depends on the chosen prior. (ii) The MCUB construction instead takes the union of projections over $\mathcal{D}$, and its frequentist coverage holds* uniformly *over the true $\rho_{DU} \in \mathcal{D}$, that is, without weighting values within $\mathcal{D}$. This paper follows the second route. The literature on imprecise probability and robust Bayes (Walley, 1991; Augustin et al., 2014) studies objects of this form, such as credal sets and lower and upper probabilities, and treats $\mathcal{D}$ as a credal family of priors rather than a single prior. From that viewpoint, Corollary 3 is a sensitivity statement for the* upper coverage probability $\sup_{r\in\mathcal{D}} \Pr_{\rho_{DU}=r}(0 \in \hat{\mathcal{C}}_{MCUB})$. *The "robustly valid" criterion of Section 4.4 then amounts to requiring coverage under two members of this family within the same structurally admissible half: a canonical member and a wider one.*

Proof of Corollary 3

Part (a). Domain monotonicity gives $\widehat{C}_n(\mathcal{D}_1) \subset \widehat{C}_n(\mathcal{D}_2)$ almost surely whenever $\mathcal{D}_1 \subset \mathcal{D}_2$, so $\{0 \in \widehat{C}_n(\mathcal{D}_1)\} \subset \{0 \in \widehat{C}_n(\mathcal{D}_2)\}$ almost surely. Taking probabilities gives Part (a).

Part (b). Since $\overline{\mathcal{D}}$ is compact and g is continuous, $C = g(\rho_0, \overline{\mathcal{D}})$ is compact. Because $0 \notin C$, compactness gives $\kappa := d(0, C) > 0$. Observe that

$$\{0 \in \widehat{C}_n(\mathcal{D})\} \ \subset\ \{d_H(\widehat{C}_n(\mathcal{D}), C) \geq \kappa\}, \tag{3}$$

since, if $0 \in \widehat{C}_n(\mathcal{D})$, then $d_H(\widehat{C}_n(\mathcal{D}), C) \geq \sup_{\hat{c}\in\widehat{C}_n(\mathcal{D})} d(\hat{c}, C) \geq d(0, C) = \kappa$. Hence

$$\mathrm{CP}_n(\delta, \mathcal{D}) = \Pr_\delta(0 \in \widehat{C}_n(\mathcal{D})) \leq \Pr_\delta\big(d_H(\widehat{C}_n(\mathcal{D}), C) \geq \kappa\big) \to 0,$$

where the limit follows from Hausdorff consistency (Lemma 6), which gives $d_H(\widehat{C}_n(\mathcal{D}), C) \to_p 0$.

Part (c). Under the unique-zero assumption, $g(\rho_0, \cdot)$ has a unique zero at $r = \rho_{DU}$. If $\rho_{DU} \notin \overline{\mathcal{D}}$, then $g(\rho_0, r) \neq 0$ for all $r \in \overline{\mathcal{D}}$, so $0 \notin C(\rho_0, \overline{\mathcal{D}})$. Part (b) therefore applies and yields $\mathrm{CP}_n(0, \mathcal{D}) \to 0$. The final sentence follows because the same argument applies to any $\mathcal{D}$ satisfying $0 \notin C(\rho_0, \overline{\mathcal{D}})$, whether ρ_{DU} is excluded by sign misspecification or by a domain that fails to contain ρ_{DU}. ■

C.4. Proof of Proposition 4

Proof *Part (a).* Write $a := \rho_{DY}\rho_{DZ} - \rho_{YZ}$. For $r \in (-1, 1)$, $g(\boldsymbol{\rho}, r) = 0$ if and only if $\rho_{DZ}\, r = a\sqrt{1-r^2}/\sqrt{1-\rho_{DY}^2}$. Dividing by $\rho_{DZ}\sqrt{1-r^2} \neq 0$ gives $r/\sqrt{1-r^2} = k$. The map $r \mapsto r/\sqrt{1-r^2}$ is a strictly increasing bijection from $(-1, 1)$ onto $\mathbb{R}$, so the zero exists, is unique, and equals $k/\sqrt{1+k^2}$. Because $C(\rho, \overline{\mathcal{D}}) = \{g(\boldsymbol{\rho}, r) : r \in \overline{\mathcal{D}}\}$, zero belongs to it if and only if $g(\boldsymbol{\rho}, r) = 0$ for some $r \in \overline{\mathcal{D}}$, that is, if and only if $r^* \in \overline{\mathcal{D}}$.

Part (b). Correlations do not change when the variables are rescaled, so let D, Y, and Z have unit variance. Then $\beta_{\mathrm{IV}} = \rho_{YZ}/\rho_{DZ}$ and $\beta_{\mathrm{OLS}} = \rho_{DY}$. Let $e := Y - \beta_{\mathrm{IV}} D$. Then $\mathrm{Cov}(D, e) = \rho_{DY} - \beta_{\mathrm{IV}}$ and $\mathrm{Var}(e) = 1 - 2\beta_{\mathrm{IV}}\rho_{DY} + \beta_{\mathrm{IV}}^2$. Since $k = (\rho_{DY} - \beta_{\mathrm{IV}})/\sqrt{1-\rho_{DY}^2}$,

$$\frac{k}{\sqrt{1+k^2}} = \frac{\rho_{DY} - \beta_{\mathrm{IV}}}{\sqrt{1 - \rho_{DY}^2 + (\rho_{DY} - \beta_{\mathrm{IV}})^2}} = \frac{\mathrm{Cov}(D, e)}{\sqrt{\mathrm{Var}(e)}} = \mathrm{cor}(D, e).$$

In these units, $\beta_{\mathrm{OLS}} - \beta_{\mathrm{IV}} = \mathrm{Cov}(D, e) = r^* \,\mathrm{sd}(e)$. In the original units, both slopes scale by $\mathrm{sd}(Y)/\mathrm{sd}(D)$ and $\mathrm{sd}(e)$ scales by $\mathrm{sd}(Y)$, which gives the stated identity. The algebra uses only first and second moments, so it also holds for sample moments.

Part (c). For a valid instrument, the IV estimand $\mathrm{Cov}(Z, Y)/\mathrm{Cov}(Z, D)$ is a weighted average of the individual effects, which we denote β_{LATE}. Hence $\mathrm{Cov}(Z, U) = \mathrm{Cov}(Z, Y) - \beta_{\mathrm{LATE}}\,\mathrm{Cov}(Z, D) = 0$, so $\rho_{ZU} = 0$, and part (b) gives $r^* = \mathrm{cor}(D, U) = \rho_{DU}$. Writing $Y = \beta_i D + U_0$, with U_0 the structural error, gives $U = (\beta_i - \beta_{\mathrm{LATE}})D + U_0$, so $\mathrm{Cov}(D, U) = \mathrm{Cov}(D, U_0) + \mathrm{Cov}\big(D, (\beta_i - \beta_{\mathrm{LATE}})D\big)$ mixes confounding and effect variation. ■

Appendix D. Simulation DGP Specifications

D.1. Synthetic benchmark (Jones-Pewsey DGP)

The synthetic benchmark of Section 5.1 uses the Jones-Pewsey sinh-arcsinh family (Jones & Pewsey, 2026) to generate a non-Gaussian regressor and errors with controlled endogeneity.

Endogenous regressor. Draw $\nu \sim N(0, 1)$ and apply the J-P transformation $T(x; \zeta, \omega) = \sinh\big[\omega \sinh^{-1}(x) - \zeta\big]$ with $\zeta = 0$ and $\omega = 0.9$. Then add an independent draw $S(\,\cdot\,;\, 0,\, 1.2)$, where $S(\cdot; \zeta, \omega)$ samples from the J-P sinh-arcsinh distribution: it takes $\xi \sim N(0, 1)$ and returns $\sinh\big((\sinh^{-1}(\xi) + \zeta)/\omega\big)$. The regressor is

$$X = 5\, T(\nu;\, 0,\, 0.9) + S(\,\cdot\,;\, 0,\, 1.2).$$

Both terms are symmetric and unimodal, so X is symmetric and unimodal, with slightly lighter tails than a normal variable (excess kurtosis -0.18).

Structural error. The endogenous component is $e = \lambda(X - \bar{X}) + \varepsilon$ with $\varepsilon \sim N(0, \sigma_X^2)$, where $\sigma_X = \mathrm{sd}(X)$ in the realised sample; the sign of λ sets the sign of ρ_{XU}. To build the exogenous component, we draw $u_1 = \mathrm{Uniform}(-1, 1) + S(\cdot\,;\, 0,\, 1.2)$, take the residual $v = u_1 - \widehat{\mathrm{lm}}(u_1 \sim X)$, and rescale

$$v \;\leftarrow\; v \cdot \frac{\mathrm{sd}(X)/2}{\mathrm{sd}(v)},$$

so that $\mathrm{sd}(v) \approx \mathrm{sd}(X)/2$. The total error is $U = e + v$, which ensures $\mathrm{cov}(X, U) = \lambda \,\mathrm{var}(X)$, positive when $\lambda > 0$ and negative when $\lambda < 0$. The outcome is $Y = \beta X + U$ with $\beta = 2$. We calibrate $|\lambda|$ by a short bisection so that $\rho_{XU} \approx \pm 0.30$, which represents substantial but not extreme endogeneity.

Instrument construction. We build eleven instruments at known ρ_{ZU} targets with a Cholesky-style projection on (X, U). For targets $a = \mathrm{cor}(Z, X)$ and $b = \mathrm{cor}(Z, U)$, with $\rho = \mathrm{cor}(X, U)$:

1. Standardise X, U; orthogonalise $U_\perp = (U - \rho X)/\sqrt{1 - \rho^2}$.
2. Draw $W \sim N(0, 1)$ and orthogonalise against $(X, U_\perp)$, then standardise.
3. Set $Z = c_1 X + c_2 U_\perp + c_3 W$ with $c_1 = a$, $c_2 = (b - a\rho)/\sqrt{1 - \rho^2}$, $c_3 = \sqrt{1 - c_1^2 - c_2^2}$.

We use $a = 0.25$ throughout, so that first-stage strength is matched, and $b \in \{-0.20, -0.16, -0.12, -0.08, -0.04, 0, +0.04, +0.08, +0.12, +0.16, +0.20\}$ across the eleven instruments.

Domain $\mathcal{D}$ in the simulations. Following the protocol of Section 4.4, we draw instruments with $b < 0$ from the $\lambda < 0$ DGP and test them under $\mathcal{D} = (-0.8,\, 0)$; we draw instruments with $b \geq 0$ from the $\lambda > 0$ DGP and test them under $\mathcal{D} = (0,\, 0.8)$. The outer bound $|0.8|$, rather than 1, avoids numerical instability near the boundary of the correlation space.

Heteroskedastic valid instrument. The row Z_{het} of Table 1 uses its own sample of $n = 800$. We draw $Z = 5\,T(\nu;\, 0,\, 0.9) + S(\cdot\,;\, 0,\, 1.2)$ and a base error $u_0 = \mathrm{Uniform}(-1, 1) + S(\cdot\,;\, 0,\, 1.2)$, independent of Z, and standardise both. The structural error is $U = s(Z)\, u_0$ with $s(Z) = \sqrt{(1 + Z^2)/2}$, so that $\mathbb{E}[s(Z)^2] = 1$. Then $\mathbb{E}[U \mid Z] = s(Z)\,\mathbb{E}[u_0] = 0$, so Z is valid, but $\mathrm{Var}(U \mid Z) = s(Z)^2$ changes with Z, so Z and U are dependent. The treatment is $X = \pi Z + r u_0 + \sqrt{1 - r^2}\, w$ with $w \sim N(0, 1)$ and $\pi = 0.25/\sqrt{1 - 0.25^2}$, which gives $\mathrm{cor}(Z, X) = 0.25$; we choose r by root-finding so that $\mathrm{cor}(X, U) = 0.30$. The outcome is $Y = 2X + U$, and the test uses $\mathcal{D} = (0,\, 0.8)$. Unlike the eleven grid instruments, Z_{het} does not have an exact sample correlation with U. In the $n = 800$ sample, this correlation is 0.083, which accounts for $\hat{\beta}_{\text{IV}} = 2.325$ in Table 1.

Across the grid of Table 1, the IV bias, $\hat{\beta}_{\text{IV}} - 2$, grows linearly in ρ_{ZU} with slope 4.6 for $\rho_{ZU} > 0$ and 4.9 for $\rho_{ZU} < 0$, close to the population value $\mathrm{sd}(U)/(\mathrm{cor}(Z, X)\,\mathrm{sd}(X)) \approx 4.7$, which confirms the DGP construction.

D.2. Structural simulation (Section 5.2)

The DGP is

$$\begin{aligned} Y &= \beta_i D + \gamma Z + \boldsymbol{\phi}^\top H + U, & V &= 0.40\, C + \varepsilon_V, \\ D &= \pi_i Z + \boldsymbol{\lambda}^\top H + V, & U &= 0.40\, C + \varepsilon_U, \end{aligned}$$

with $Z, C \sim \mathcal{N}(0,1)$, $H \sim \mathcal{N}(0, I_5)$, and $\varepsilon_V, \varepsilon_U \sim \mathcal{N}(0,\ 0.6^2)$, all independent. Each unit belongs to one of two latent types with probability $1/2$, and the type sets (β_i, π_i). The loading 0.40 on C gives $\mathrm{cor}(V,U) = 0.16/0.52 \approx 0.31$.

We draw each element of the coefficient vectors $\boldsymbol{\phi}, \boldsymbol{\lambda} \in \mathbb{R}^5$ i.i.d. $\mathcal{N}(0,1)$, once, under `set.seed(2026)`, and save them to `coef_vectors.rds` so that all runs use the same values. We write $\boldsymbol{\phi}$ rather than the more conventional $\boldsymbol{\delta}$ to avoid a clash with the scalar exclusion-violation magnitude $\delta := |\rho_{ZU}|$ defined in Section 4.

Each row of Table 9 draws its own sample of $n = 50{,}000$, except row B (conf.), which reuses the sample of row A.

- Row A: $\beta_i = 0.425$ and $\pi_i = 0.25$ for both types, and $\gamma = 0$; the analyst sees (H, C). IV and DML both target 0.425.
- Row B (het.): $(\beta_i, \pi_i) = (0.80, 0.05)$ or $(0.05, 0.60)$, and $\gamma = 0$; the analyst sees (H, C). The ATE is 0.425. IV targets $\beta_{\text{LATE}} = \mathbb{E}[\beta_i \pi_i]/\mathbb{E}[\pi_i] = 0.108$. The partially linear DML estimand weights each type by its residual treatment variance $w_i = \pi_i^2 + 0.6^2$, which gives $\theta_{\text{DML}} = \mathbb{E}[\beta_i w_i]/\mathbb{E}[w_i] = 0.301$.
- Row B (conf.): the sample of row A, but the analyst sees only H. IV still targets 0.425, while DML targets $0.425 + \mathrm{Cov}(V,U)/\mathrm{Var}(D \mid H) = 0.425 + 0.16/0.5825 = 0.700$.
- Row C: as row A, with $\gamma = 0.30$.
- Row C (weak): as row A, with $\pi_i = \pi_{\text{weak}} = \sqrt{5\,\sigma_V^2/n}$ and $\sigma_V^2 = 0.52$. Given C, the concentration parameter is $n\pi_{\text{weak}}^2/0.36 \approx 7$, so $F < 10$ at any n; at $n = 50{,}000$, $\pi_{\text{weak}} \approx 0.007$. Because $\gamma = 0$, this instrument is weak but valid.

H enters both the DML controls and the IV regression as an exogenous covariate, and so does C when the analyst sees it. The CR test residualises on the same covariates as DML.

D.3. Classifiability frontier values

We calibrate σ on the eight invalid empirical instruments. For each, $\hat{\sigma} = \sqrt{n}\,|\hat{\ell} - \hat{\ell}_{\text{MCUB}}|/z_{0.975}$, where $\hat{\ell}$ and $\hat{\ell}_{\text{MCUB}}$ are the plug-in and MCUB endpoints nearest zero. The eight values range from 1.69 to 6.75, with median 3.25, which Table 6 uses.

Equation (2) gives $\kappa_N^* \approx 0.036$ at $n = 1{,}682$ (MovieLens) and ≈ 0.044 at $n = 1{,}107$ (NHANES), both at $\pi^* = 0.05$. All eight invalid instruments lie beyond the frontier at their sample sizes. The closest case is `Card fatheduc`: its plug-in identified set starts 0.067 from zero at $n = 2{,}320$, about twice the corresponding $\kappa_N^* \approx 0.031$. Its MCUB set still reaches zero, and the bootstrap rule, not a single set, detects the violation ($\widehat{p}_B = 0.814$). With Card's own calibrated $\hat{\sigma} = 3.39$, the Gaussian approximation of Remark 5 predicts that a resampled set contains zero with probability $\Phi(1.96 - \sqrt{2{,}320} \times 0.067/3.39) = 0.84$, close to the observed 0.814.

Appendix E. Monte Carlo Validation of Proposition 2

We validate the finite-sample bounds over $R = 1{,}000$ independent draws from the Jones-Pewsey DGP of Appendix D.1, at $n = 800$, the sample size of Table 1. Each draw runs one CR test, with $B_{\text{mcub}} = 500$ draws per bisection step, $B_{\text{large}} = 499$ coverage draws, and a bootstrap covariance from $B_{\text{boot}} = 200$ draws that redraw the simulated rows and redo the centring and projection; it

Table 6: Classifiability frontier κ_N^* on the distance scale: the smallest distance between zero and the identified set that the bootstrap rule of Proposition 2 resolves with error at most π^*. Values come from equation (2) with $\tau_V = 0.95$, $B = 500$, $\alpha_0 = 0.05$, and $\sigma = 3.25$, the median calibrated value (Section 4.3), and they scale linearly in σ. They are not comparable with the δ-scale crossings δ_n^* of Figure 2, which sit above the population threshold $\delta_\infty^* = 0.125$.

n	$\pi^* = 0.01$	$\pi^* = 0.05$	$\pi^* = 0.10$	$\pi^* = 0.20$
1,107	0.049	0.044	0.042	0.038
1,534	0.042	0.038	0.035	0.032
5,000	0.023	0.021	0.020	0.018
10,000	0.016	0.015	0.014	0.013
50,000	0.007	0.007	0.006	0.006

uses the single-sample verdict, not the bootstrap rule (Section 4.3). The instrument grid spans $\rho_{ZU} \in \{-0.20, -0.16, -0.12, -0.08, -0.04, 0, +0.04, +0.08, +0.12, +0.16, +0.20\}$, covering both signs. Following the protocol of Section 4.4, each cell uses its correctly elicited canonical domain: cells with $\rho_{ZU} < 0$ draw from the $\lambda < 0$ DGP and test under $\mathcal{D} = (-0.8, 0)$, and cells with $\rho_{ZU} \geq 0$ draw from the $\lambda > 0$ DGP and test under $\mathcal{D} = (0, 0.8)$. Each replication's verdict applies the finite-sample equivalence margin $\text{ZEROTOL} = 0.05$ to the MCUB endpoints (footnote 2).

Table 7: Monte Carlo validation under the J-P DGP ($R = 1{,}000$ replications, $n = 800$, $B_{\text{mcub}} = 500$, $B_{\text{large}} = 499$, $B_{\text{boot}} = 200$). Column $\bar{\hat{p}}$ is the fraction of replications in which zero lies in the MCUB set $\hat{\mathcal{C}}_{\text{MCUB}}$, within tolerance. Truth labels: $|\rho_{ZU}| \leq 0.04 \Rightarrow$ A and $|\rho_{ZU}| \geq 0.16 \Rightarrow$ C. The cells with $|\rho_{ZU}| \in \{0.08, 0.12\}$ form the boundary region; they have no truth label, are not scored, and are marked *Boundary*. We report $\bar{\hat{p}}$ on these rows because it shows where the empirical detection threshold falls. The last column is the exact one-sided 95% upper confidence bound on each scored cell's misclassification rate. With $0/1000$ errors, it equals $1 - 0.05^{1/1000} = 0.003$, and with $1/1000$, it equals 0.005; both lie below the target $\pi^* = 0.05$.

Instrument	ρ_{ZU}	$\mathcal{D}$	$\bar{\hat{p}}$	DIAL	Misclass.	95% UCB
$Z_{-0.20}$	-0.20	$(-0.8, 0)$	0.000	**C**	$0/1000$	0.003
$Z_{-0.16}$	-0.16	$(-0.8, 0)$	0.001	**C**	$1/1000$	0.005
$Z_{-0.12}$	-0.12	$(-0.8, 0)$	0.980	Boundary	—	—
$Z_{-0.08}$	-0.08	$(-0.8, 0)$	1.000	Boundary	—	—
$Z_{-0.04}$	-0.04	$(-0.8, 0)$	1.000	**A**	$0/1000$	0.003
$Z_{+0.00}$	0.00	$(0, 0.8)$	1.000	**A**	$0/1000$	0.003
$Z_{+0.04}$	$+0.04$	$(0, 0.8)$	1.000	**A**	$0/1000$	0.003
$Z_{+0.08}$	$+0.08$	$(0, 0.8)$	1.000	Boundary	—	—
$Z_{+0.12}$	$+0.12$	$(0, 0.8)$	0.990	Boundary	—	—
$Z_{+0.16}$	$+0.16$	$(0, 0.8)$	0.000	**C**	$0/1000$	0.003
$Z_{+0.20}$	$+0.20$	$(0, 0.8)$	0.000	**C**	$0/1000$	0.003

The transition from Valid to Invalid occurs in a narrow band, $|\rho_{ZU}| \in (0.12, 0.16)$, symmetric about zero. Inside the scored region, $\bar{\hat{p}}$ lies within 0.001 of 0 or 1: every cell with $|\rho_{ZU}| \leq 0.04$ accepts in all 1,000 replications, and every cell with $|\rho_{ZU}| \geq 0.16$ rejects in at least 999. The boundary cells accept in every replication at $|\rho_{ZU}| = 0.08$ and in 98% and 99% of replications at $|\rho_{ZU}| = 0.12$. Linear interpolation between 0.12 and 0.16 places the empirical 50% detection threshold near $|\rho_{ZU}| \approx 0.14$. This threshold lies well above $|\rho_{XU}| \operatorname{cor}(Z, X) = 0.30 \times 0.25 = 0.075$, the violation below which the identified set contains zero. At $n = 800$, sampling error accounts for most of the difference, and the ZEROTOL margin for the rest. A single MCUB set therefore rarely detects $|\rho_{ZU}| = 0.12$ at this sample size, so we treat these cells as boundary cells. The bootstrap rule is stricter than a single set: in Table 1, it rejects both instruments at $|\rho_{ZU}| = 0.12$ ($\widehat{p}_B = 0.728$ and 0.700), although their MCUB sets contain zero. Outside the boundary region, the classification matches Table 1 exactly. The non-Gaussian regressor and errors apply to every cell, so every row of the validation builds in robustness to non-Gaussian dependence by construction.

The bootstrap rule itself. Table 7 scores a single MCUB set per sample. Table 8 runs the bootstrap rule of Algorithm 1 on three cells of the same design: $\rho_{ZU} = 0$ (valid), 0.12 (boundary), and 0.20 (invalid). Each cell reuses the first 100 samples of Table 7 and applies the rule with $B = 200$ resamples, so the single set and the rule are compared on identical data. The rule makes no error in the two scored cells. It accepts the valid instrument in all 100 replications (mean $\widehat{p}_B = 1.000$) and rejects the invalid one in all 100 (mean $\widehat{p}_B = 0.031$), so both exact 95% upper bounds equal 0.030, below $\pi^* = 0.05$. At the boundary cell, the single set contains zero in 98 replications, but the rule accepts in none of the 100: $\widehat{p}_B$ averages 0.706, and its 10th and 90th percentiles, 0.605 and 0.795, both lie below τ_V. Strict containment gives the same verdicts in all three cells. Across 100 samples, this confirms what Table 1 shows for one: the rule detects a violation of $|\rho_{ZU}| = 0.12$ that a single set misses at $n = 800$.

Table 8: The bootstrap validity rule under the J-P DGP ($n = 800$, 100 replications per cell, $B = 200$ resamples, $\alpha_0 = 0.10$). "Single set" is the share of replications whose full-sample MCUB set contains zero within ZEROTOL, and "Rule" the share with $\widehat{p}_B \geq \tau_V = 0.95$, with the margin and with strict containment. The last two columns score the rule with the margin against the truth label and give the exact one-sided 95% upper confidence bound on its misclassification rate.

ρ_{ZU}	Truth	Single set	Rule	Rule (strict)	Mean $\widehat{p}_B$	Misclass.	95% UCB
0.00	A	1.00	1.00	1.00	1.000	0/100	0.030
+0.12	Boundary	0.98	0.00	0.00	0.706	—	—
+0.20	C	0.00	0.00	0.00	0.031	0/100	0.030

Appendix F. Experimental Tables and Figures

This appendix contains Figure 4 and Table 9, which Section 5 discusses.

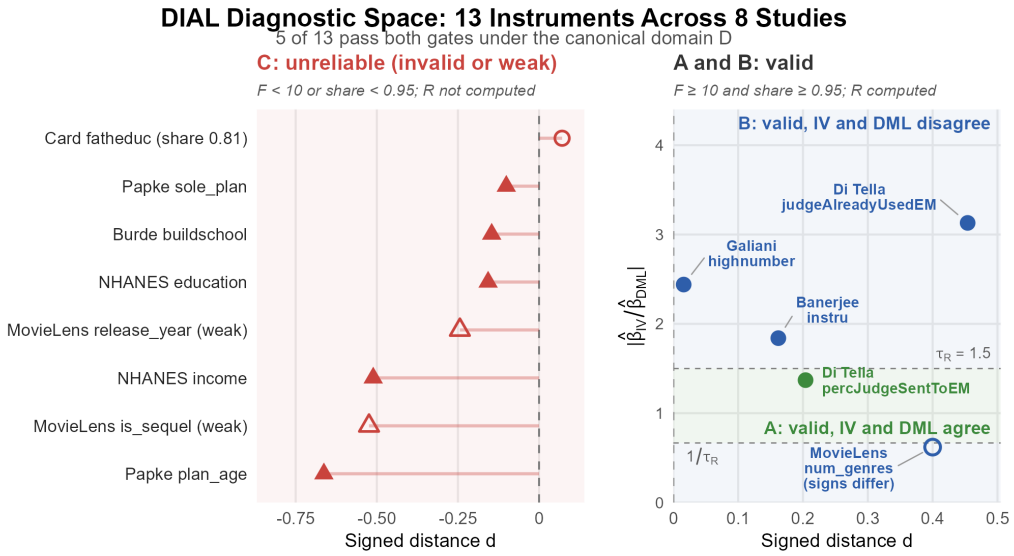


Figure 4: DIAL diagnostic space for 13 instruments across 8 published studies, each under its sign-elicited canonical domain $\mathcal{D}$ (Section 4.4). Both panels use the signed distance d from zero to the nearest endpoint of $\hat{\mathcal{C}}_{\text{MCUB}}$, computed from the endpoints in Table 3: $d > 0$ when $0 \in \hat{\mathcal{C}}_{\text{MCUB}}$ and $d < 0$ otherwise. **Left:** the eight Scenario C instruments (unreliable: invalid or weak). For these, Algorithm 1 stops at the relevance or validity gate, so it does not compute R, and the panel places them by d alone. Open triangles mark the two weak instruments, whose MCUB sets also exclude zero. An open circle marks Card: its set contains zero ($d > 0$), but the bootstrap rule rejects it ($\widehat{p}_B = 0.814$). **Right:** the five instruments that pass both gates, placed by d and R. Dashed lines at $R = 1/\tau_R$ and $R = \tau_R$, with $\tau_R = 1.5$, bound Scenario A (IV and DML agree); Scenario B (IV and DML disagree) lies outside this band. An open circle marks MovieLens `num_genres`, whose IV and DML estimates have opposite signs. Under each instrument's sign-elicited canonical $\mathcal{D}$, 5 of the 13 instruments pass the validity gate (one A, four B), and 8 fail it; no verdict changes under within-sign widening (Table 4). Di Tella supplies both an A case and a B case in the same dataset (Section 5.3). Table 5 gives bootstrap intervals for R.

Table 9: Structural simulation: five analyses, one per row, that cover all three DIAL scenarios. Each row has its own DML fit and, except row B (conf.), its own sample of $n = 50{,}000$. Population targets: IV and DML 0.425 in rows A and C (weak); IV 0.108 and DML 0.301 in row B (het.); IV 0.425 and DML 0.700 in row B (conf.), where the analyst does not see the confounder C. In row C, the exclusion violation biases IV away from the effect of 0.425. $R = |\hat{\beta}_{\text{IV}}|/|\hat{\beta}_{\text{DML}}|$; $\widehat{p}_B$ is the share of $B = 500$ bootstrap resamples whose MCUB set contains zero within ZEROTOL $= 0.05$; and r^* is the implied endogeneity of Proposition 4.

Analysis	$\hat{\beta}_{\text{DML}}$	$\hat{\beta}_{\text{IV}}$	R	F	$\hat{\mathcal{C}}_{\text{MCUB}}$	$\widehat{p}_B$	r^*	DIAL
A: homogeneous	0.444	0.429	0.96	8,814	$[-0.009, +0.810]$	1.000	-0.004^*	**A**
B (het.): LATE $< \theta_{\text{DML}}$	0.314	0.109	0.35	12,575	$[-0.120, +0.737]$	1.000	$+0.109$	**B**
B (conf.): C unobserved	0.670	0.432	0.64	6,199	$[-0.296, +0.610]$	1.000	$+0.282$	**B**
C: exclusion ($\gamma = 0.30$)	0.601	1.658	2.76	8,733	$[+0.707, +1.000]^\dagger$	0.000	-0.714	**C** (invalid)
C (weak): $\pi = 0.007$	0.446	0.281	0.63	7	$[-0.740, +1.000]^\dagger$	0.974	$+0.144$	**C** (weak)

* In row A, the population r^* is 0, the lower edge of $\mathcal{D} = (0, 0.8)$, so the sample value can fall just outside $\mathcal{D}$. The MCUB width and ZEROTOL absorb this sampling noise (footnote 2).
† The MCUB upper endpoints exceed 1 before clipping ($+1.002$ and $+1.167$); we report $+1.000$ because $\rho_{ZU} \in [-1, 1]$. The verdicts are unaffected.

Appendix G. Implementation and Reproducibility Details

Code. Replication code for every table and figure is at
https://anonymous.4open.science/r/dial-cr-replication-0B38.
The scripts in `scripts/` are numbered `00`–`10`, and each one produces specific tables or figures from raw inputs. Two shared files implement the CR test with MCUB inference and the bootstrap validity rule.

Estimation settings. The DML nuisance functions use random forests with 5-fold cross-fitting, 500 trees (300 in the structural simulation), and a minimum node size of 5, following the standard specification of Chernozhukov et al. (2018); the empirical applications repeat the sample split three times. The CR test runs on a prepared frame: we residualise Y, D, and Z on the covariates (on an intercept when there are none) and replace the residualised instrument by its fitted value from a regression on the residualised (D, Y). Every bootstrap in the pipeline redraws the rows of the analysis data and redoes this preparation. The fitted values depend on estimated coefficients, and resampling the prepared rows would hold those coefficients fixed and understate the sampling error of the correlations. Each full-sample MCUB set uses 500 draws per bisection step, 5,000 coverage draws, and a bootstrap covariance of the correlations from 800 such draws, at the nominal level $\alpha_0 = 0.05$ in the empirical applications and the power curves and 0.10 in the other simulations. The validity rule draws $B = 500$ bootstrap resamples and recomputes the MCUB set on each, with 500 bisection draws, 499 coverage draws, and a covariance from 200 draws of the resample; it uses the threshold $\tau_V = 0.95$ and seed $5000 + i$ for instrument index i. The intervals for r^* use 2,000 resamples of the raw rows. Those for R use 200, refit DML on each resample, and assign the cross-fitting folds to the original rows, so that copies of a row never sit in both the training and the test fold. Because the MCUB critical values come from simulation, rounding differences between computers can move set endpoints in the third decimal and change $\widehat{p}_B$ by a few resamples.

The domain $\mathcal{D}$ is set for each instrument by the sign-elicitation protocol of Section 4.4. The ratio threshold $\tau_R = 1.5$ is a practical convention: inside $(1/\tau_R, \tau_R)$, neither estimate exceeds the other by more than 50%. The Monte Carlo validation uses $n = 800$ and $R = 1{,}000$ replications (Appendix E). The power curves use a 43-point δ grid (35 base points in $[0.001, 0.40]$ plus 8 extra points that resolve the transition region) across five sample sizes, $n \in \{1{,}107,\ 1{,}534,\ 5{,}000,\ 10{,}000,\ 50{,}000\}$, with $R \in \{250, 250, 150, 100, 50\}$ replications per cell, respectively. They use $B_{\text{mcub}} = 500$ bisection-iteration draws, $B_{\text{large}} = 499$ coverage draws, and a bootstrap covariance estimated from $B_{\text{boot}} = 200$ draws. The replication script records the seed that draws and fixes the coefficient vectors $\boldsymbol{\phi}$ and $\boldsymbol{\lambda}$ of the structural simulation.

Data. All empirical datasets are public, and Section 5.3 cites each one where it is used. We use MovieLens 100K under the GroupLens Research dataset terms (Harper & Konstan, 2015). NHANES is a public-use U.S. federal dataset. Its analytic frame is built by `scripts/01_nhanes_build.R`, which reads `data(nhanes.fish)` from the `CrossScreening` R package, the same 1,107-row extract that Rosenbaum (2014) and Zhao et al. (2018) document. The Papke and Card data come from the `wooldridge` R package (GPL-3). The Di Tella, Galiani, Banerjee, and Burde & Linden data come from their journal replication archives, and the repository includes the four `.dta` files. All packages are used in accordance with their open-source licences.

Compute. All experiments run on a standard laptop or workstation CPU and need no GPU; peak memory is under 4 GB. The synthetic benchmark (Table 1) takes about 45 minutes on 7 cores, including the bootstrap rule. The Monte Carlo validation (Appendix E) takes about three hours on 7 cores with `future.apply`. The classifiability-frontier grid (Figure 2) takes about 160 minutes on 7 cores. The structural simulation (Table 9) takes about 40 minutes on 7 cores, including the bootstrap rule. The implied endogeneity and the intervals for R (Table 5) take about 20 minutes, and the Monte Carlo of the bootstrap rule (Table 8) about eight and a half hours. The empirical applications (Tables 3 and 4) take about 2 hours in total on 7 cores, of which the bootstrap rule takes about 70 minutes; DML random-forest nuisance estimation and the `CIhybrid` bisection loop dominate the runtime.